

# LAPORAN DATA CLEANING

## E-Commerce Sales Dataset — Day 1

Roadmap 7 Hari Data Analyst Portfolio — Dokumentasi Lengkap Langkah demi Langkah

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### 1. Pendahuluan

Laporan ini mendokumentasikan secara rinci setiap langkah proses pembersihan data (data cleaning) pada dataset transaksi e-commerce (raw\_sales\_data.csv) menggunakan Microsoft Excel, sebagai bagian dari Day 1 dalam roadmap pengembangan portofolio Data Analyst. Dataset mentah berisi 4.100 baris transaksi periode Januari 2023 – Desember 2024, dengan sejumlah masalah kualitas data yang sengaja merepresentasikan kondisi data dunia nyata: duplikasi, nilai kosong, inkonsistensi format, kesalahan tipe data, dan outlier.

Tujuan tahap ini adalah menghasilkan dataset yang bersih, konsisten, dan siap dipakai pada tahap selanjutnya (analisis Excel, SQL, Python, dan dashboard Tableau), sehingga angka yang dilaporkan pada seluruh tahap roadmap konsisten satu sama lain. Setiap langkah di bawah ini disertai formula yang dipakai dan tangkapan layar (screenshot) sebagai bukti proses, bukan hanya ringkasan naratif, sehingga dapat dijadikan referensi ulang tanpa perlu mengingat dari nol.

### 2. Ringkasan Eksekutif

Tabel 1 merangkum perubahan kondisi dataset sebelum dan sesudah proses cleaning.

| Metrik                             | Sebelum         | Sesudah               |
|------------------------------------|-----------------|-----------------------|
| Jumlah baris                       | 4.100           | 3.933                 |
| Jumlah kolom                       | 11              | 12 (+ revenue)        |
| Baris duplikat                     | 99              | 0                     |
| Missing value (seluruh kolom)      | 533 sel         | 0                     |
| Outlier ekstrem (quantity & harga) | 27 baris        | 0                     |
| Format order_date                  | 5 format campur | 1 format (YYYY-MM-DD) |

Tabel 1. Perbandingan kondisi dataset sebelum dan sesudah cleaning

### 3. Tahapan Data Cleaning

#### 3.1 Penghapusan Data Duplikat

Baris duplikat diidentifikasi menggunakan fitur Data > Remove Duplicates dengan seluruh kolom dicentang, sehingga hanya baris yang identik persis di semua kolom (termasuk order\_id) yang dianggap duplikat — bukan sekadar kemiripan sebagian data. Pendekatan ini menghindari kesalahan menghapus dua transaksi berbeda yang kebetulan memiliki produk atau tanggal yang sama.

## Langkah 1 — Identifikasi baris duplikat dengan row signature

Untuk memastikan hanya baris yang benar-benar identik yang tersorot, seluruh kolom digabung menjadi satu "tanda tangan baris" (row signature) memakai TEXTJOIN, lalu tanda tangan yang identik disorot dengan Conditional Formatting > Highlight Duplicate Values:

```
=TEXTJOIN(" | ", TRUE, A2:K2)
```

|    | D              | E        | F                 | G                      | H        | I          | J            | K              | L                                                                                                                           |
|----|----------------|----------|-------------------|------------------------|----------|------------|--------------|----------------|-----------------------------------------------------------------------------------------------------------------------------|
|    | customer_name  | region   | category          | product_name           | quantity | unit_price | order_status | payment_method | row_signature                                                                                                               |
| 1  | Andi Halim     |          | Sports & Outdoors | Water Bottle           | 2        | 28.97      | Completed    | Credit Card    | ORD10016145253   CUST11353   Andi Halim   Sports & Outdoors   Water Bottle   2   28.97   Completed   Credit Card            |
| 2  | Lisa Kurniawan | East     | Toys              | Building Blocks        | 2        | 46.3       | Returned     | Credit Card    | ORD10279414499   CUST14781   Lisa Kurniawan   East   Toys   Building Blocks   2   46.3   Returned   Credit Card             |
| 3  | Ahmad Halim    | Mid-West | Beauty            | Moisturizer            | 1        | 35.09      | Completed    | Gift Card      | ORD10349045392   CUST15312   Ahmad Halim   Mid-West   Beauty   Moisturizer   1   35.09   Completed   Gift Card              |
| 4  | Ayu Hidayat    | SOUTH    | Sports & Outdoors | Camping Tent           | 2        | 12.94      | Completed    | Debit Card     | ORD10156745526   CUST13901   Ayu Hidayat   SOUTH   Sports & Outdoors   Camping Tent   2   12.94   Completed   Debit Card    |
| 5  | James Yusuf    | SOUTH    | sports & outdoors | Hiking Backpack        | 1        | 23.12      | Completed    | PayPal         | ORD10069645343   CUST15853   James Yusuf   SOUTH   sports & outdoors   Hiking Backpack   1   23.12   Completed   PayPal     |
| 6  | Fajar Halim    | Midwest  | Beauty            | Sunscreen SPF50        | 1        | 33.88      | Completed    | Gift Card      | ORD10147345522   CUST10521   Fajar Halim   Midwest   Beauty   Sunscreen SPF50   1   33.88   Completed   Gift Card           |
| 7  | John Tan       | south    | books             | Fiction Novel          | 2        | 10.88      | Completed    | Gift Card      | ORD10311145519   CUST17261   John Tan   south   books   Fiction Novel   2   10.88   Completed   Gift Card                   |
| 8  | Kevin Anderson | WEST     | Beauty            | Moisturizer            | 1        | 21.46      | Completed    | PayPal         | ORD10053745119   CUST14431   Kevin Anderson   WEST   Beauty   Moisturizer   1   21.46   Completed   PayPal                  |
| 9  | John Yusuf     | MIDWEST  | Home & Kitchen    | Cutting Board          | 1        | 102.66     | Completed    | PayPal         | ORD10043145048   CUST16991   John Yusuf   MIDWEST   Home & Kitchen   Cutting Board   1   102.66   Completed   PayPal        |
| 10 | Ayu Smith      | South    | sports & outdoors | Resistance Bands       | 1        | 56.4       | Completed    | PayPal         | ORD101315145149   CUST17081   Ayu Smith   South   sports & outdoors   Resistance Bands   1   56.4   Completed   PayPal      |
| 11 | Nina Lee       | midwest  | Books             | Business Strategy Book | 1        | 18.34      | Completed    | Credit Card    | ORD10276745191   CUST18881   Nina Lee   midwest   Books   Business Strategy Book   1   18.34   Completed   Credit Card      |
| 12 | Ayu Hidayat    | south    | Clothing          | Winter Jacket          | 2        | 35.48      | Completed    | Gift Card      | ORD10188845366   CUST12111   Ayu Hidayat   south   Clothing   Winter Jacket   2   35.48   Completed   Gift Card             |
| 13 | Fajar Santos   | east     | Home & Kitchen    | Storage Containers     | 1        | 112.68     | Completed    | Gift Card      | ORD10074945500   CUST11341   Fajar Santos   east   Home & Kitchen   Storage Containers   1   112.68   Completed   Gift Card |
| 14 | Sarah Tan      | East     | Electronics       | Webcam                 | 1        | 68.92      | Completed    | Debit Card     | ORD10185745143   CUST11041   Sarah Tan   East   Electronics   Webcam   1   68.92   Completed   Debit Card                   |
| 15 | Lisa Taylor    | MIDWEST  | Beauty            | Shampoo                | 2        | 23.18      | Cancelled    |                | ORD10080045114   CUST13321   Lisa Taylor   MIDWEST   Beauty   Shampoo   2   23.18   Cancelled                               |
| 16 | Budi Kurniawan | West     | Clothing          | Hoodie                 | 2        | 66.21      | Returned     | PayPal         | ORD10169545432   CUST12251   Budi Kurniawan   West   Clothing   Hoodie   2   66.21   Returned   PayPal                      |
| 17 | Linda Chen     | east     | Sports & Outdoors | Camping Tent           | 1        |            | Cancelled    | PayPal         | ORD10209345116   CUST18261   Linda Chen   east   Sports & Outdoors   Camping Tent   1                                       |
| 18 | Siti Anderson  | East     | Electronics       | Power Bank             | 3        | 134.85     | Completed    | PayPal         | ORD10129645571   CUST10031   Siti Anderson   East   Electronics   Power Bank   3   134.85   Completed   PayPal              |
| 19 | Siti Wilson    | midwest  | Sports & Outdoors | Yoga Mat               | 1        | 75.34      | Completed    | Credit Card    | ORD10186245380   CUST17411   Siti Wilson   midwest   Sports & Outdoors   Yoga Mat   1   75.34   Completed   Credit Card     |
| 20 | Nina Garcia    | SOUTH    | electronics       | Power Bank             | 5        | 126.80     | Completed    | Gift Card      | ORD101291145225   CUST18561   Nina Garcia   SOUTH   electronics   Power Bank   5   126.80   Completed   Gift Card           |
| 21 | Nina Garcia    | West     | Sports & Outdoors | Hiking Backpack        | 1        | 49.73      | Returned     | Credit Card    | ORD10295445157   CUST12301   Nina Garcia   West   Sports & Outdoors   Hiking Backpack   1   49.73   Returned   Credit Card  |
| 22 | Robert Smith   | west     | clothing          | Denim Jeans            | 4        | 62.83      | Completed    | Debit Card     | ORD10182445495   CUST13241   Robert Smith   west   clothing   Denim Jeans   4   62.83   Completed   Debit Card              |
| 23 | Siti Tan       | EAST     | Beauty            | Face Mask Set          | 1        | 39.05      | Completed    | Debit Card     | ORD10167645294   CUST16891   Siti Tan   EAST   Beauty   Face Mask Set   1   39.05   Completed   Debit Card                  |
| 24 | David Wijaya   | East     | Home & Kitchen    | Blender                | 4        | 115.34     | Completed    | Debit Card     | ORD10177745054   CUST12271   David Wijaya   EAST   Home & Kitchen   Blender   4   115.34   Completed   Debit Card           |
| 25 | David Anderson | South    | sports & outdoors | Camping Tent           | 1        | 30.3       | Completed    | Debit Card     | ORD10222445051   CUST15881   David Anderson   South   sports & outdoors   Camping Tent   1   30.3   Completed   Debit Card  |
| 26 | Sarah Johnson  | south    | Clothing          | Hoodie                 | 1        | 46.19      | Completed    | Debit Card     | ORD10181545477   CUST12651   Sarah Johnson   south   Clothing   Hoodie   1   46.19   Completed   Debit Card                 |
| 27 | Dewi Sari      | EAST     | Books             | Self-Help Guide        | 1        | 25.66      | Completed    | Credit Card    | ORD10181945496   CUST13431   Dewi Sari   EAST   Books   Self-Help Guide   1   25.66   Completed   Credit Card               |
| 28 | Putri Halim    | East     | SPORTS & OUTDOORS | Camping Tent           | 1        | 72.43      | Cancelled    | PayPal         | ORD10075645124   CUST16801   Putri Halim   East   SPORTS & OUTDOORS   Camping Tent   1   72.43   Cancelled   PayPal         |
| 29 | Maria Wilson   | SOUTH    | TOYS              | Board Game             | 1        | 58.15      | Completed    | PayPal         | ORD10191045024   CUST18871   Maria Wilson   SOUTH   TOYS   Board Game   1   58.15   Completed   PayPal                      |
| 30 | Dewi Tan       | east     | Beauty            | Lip Balm               | 2        | 7.83       | Completed    | PayPal         | ORD10032645480   CUST11821   Dewi Tan   east   Beauty   Lip Balm   2   7.83   Completed   PayPal                            |
| 31 | Nina Garcia    | MIDWEST  | TOYS              | RC Car                 | 1        | 9.37       | Completed    |                | ORD10323845032   CUST12541   Nina Garcia   MIDWEST   TOYS   RC Car   1   9.37   Completed                                   |
| 32 | Sarah Lee      | East     | Clothing          | Cotton T-Shirt         | 1        | 54.55      | Completed    | Debit Card     | ORD10263445070   CUST17571   Sarah Lee   East   Clothing   Cotton T-Shirt   1   54.55   Completed   Debit Card              |
| 33 | Rina Taylor    | SOUTH    | Beauty            | Moisturizer            | 1        | 25.47      | Completed    | Debit Card     | ORD10161645348   CUST11281   Rina Taylor   SOUTH   Beauty   Moisturizer   1   25.47   Completed   Debit Card                |
| 34 | Ahmad Hidayat  | Mid-West | Home & Kitchen    | Coffee Maker           | 2        | 52.90      | Cancelled    | Gift Card      | ORD10159945145   CUST13061   Ahmad Hidayat   Mid-West   Home & Kitchen   Coffee Maker   2   52.90   Cancelled   Gift Card   |
| 35 | Siti Wilson    | WEST     | Beauty            | Shampoo                | 2        | 54.80      | Completed    | PayPal         | ORD10122645542   CUST12061   Siti Wilson   WEST   Beauty   Shampoo   2   54.80   Completed   PayPal                         |
| 36 | Sarah Chen     | WEST     | Toys              | Board Game             | 1        | 58.26      | Completed    | Debit Card     | ORD10250745192   CUST16431   Sarah Chen   WEST   Toys   Board Game   1   58.26   Completed   Debit Card                     |
| 37 | Rina Anderson  | WEST     | Toys              | Puzzle 1000pc          | 1        | 25.48      | Completed    | Debit Card     | ORD10085744943   CUST13701   Rina Anderson   WEST   Toys   Puzzle 1000pc   1   25.48   Completed   Debit Card               |
| 38 | Budi Chen      | South    | ELECTRONIC        | Bluetooth Speaker      | 2        | 93.44      | Completed    | PayPal         | ORD101423145047   CUST18381   Budi Chen   South   ELECTRONIC   Bluetooth Speaker   2   93.44   Completed   PayPal           |
| 39 | Kevin Lee      | East     | clothing          | Running Shorts         | 2        | 24.84      | Completed    | PayPal         | ORD10033345560   CUST17531   Kevin Lee   East   clothing   Running Shorts   2   24.84   Completed   PayPal                  |
| 40 | Nina Garcia    | West     | clothing          | Hoodie                 | 2        |            | Completed    | PayPal         | ORD102984171092023   CUST16151   Putri Anderson   West   clothing   Hoodie   2                                              |
| 41 | Ayu Chen       | South    | ELECTRONIC        | Power Bank             | 2        | 142.63     | Completed    | Debit Card     | ORD10061945464   CUST14141   Ayu Chen   South   ELECTRONIC   Power Bank   2   142.63   Completed   Debit Card               |
| 42 | Dewi Lee       | midwest  | Beauty            | Sunscreen SPF50        | 2        | 15.43      | Completed    | Debit Card     | ORD10120045332   CUST16081   Dewi Lee   midwest   Beauty   Sunscreen SPF50   2   15.43   Completed   Debit Card             |
| 43 | Ayu Hidayat    | Mid-West | Clothing          | Running Shorts         | 2        |            | Completed    | Debit Card     | ORD10167545531   CUST17641   Ayu Hidayat   Mid-West   Clothing   Running Shorts   2                                         |
| 44 | Maria Anderson | South    | Toys              | Puzzle 1000pc          | 1        | 19.86      | Completed    | Gift Card      | ORD10181045069   CUST12671   Maria Anderson   South   Toys   Puzzle 1000pc   1   19.86   Completed   Gift Card              |
| 45 | Ayu Pratama    | east     | Sports & Outdoors | Yoga Mat               | 1        | 38.17      | Completed    | PayPal         | ORD1012311241032024   CUST10731   Ayu Pratama   east   Sports & Outdoors   Yoga Mat   1   38.17   Completed   PayPal        |
| 46 | Sarah Tan      | MIDWEST  | Clothing          | Cotton T-Shirt         | 4        | 38.53      | Returned     |                | ORD10296844883   CUST13011   Sarah Tan   MIDWEST   Clothing   Cotton T-Shirt   4   38.53   Returned                         |

Gambar 1. Kolom row\_signature — baris dengan tanda tangan identik tersorot warna, menandakan duplikat asli

## Langkah 2 — Remove Duplicates

Setelah baris duplikat terverifikasi, dijalankan Data > Remove Duplicates dengan seluruh kolom dicentang:

Gambar 2. Hasil Remove Duplicates — 98 baris duplikat ditemukan dan dihapus, menyisakan 4.002 baris unik

- Hasil awal: 98 baris exact-duplicate dihapus, menyisakan 4.002 baris unik.

### Langkah 3 — Verifikasi ulang (Day 2)

Sebelum dataset dipakai untuk tahap analisis, dilakukan pengecekan ulang dengan teknik yang sama (row signature + filter warna). Ditemukan 1 baris duplikat tambahan (order\_id ORD101170, customer Fajar Wijaya — Wool Socks, qty 3) yang lolos dari proses awal:

Gambar 3. Verifikasi ulang di Day 2 menemukan 1 duplikat tambahan yang lolos dari pembersihan awal (baris 441 dan 3935, isi identik)

- Total akhir: 99 baris exact-duplicate dihapus (98 pada pembersihan awal + 1 ditemukan saat verifikasi ulang).

## 3.2 Standardisasi Kolom Category & Region

Kedua kolom ini memiliki variasi kapitalisasi ("electronics", "ELECTRONICS"), spasi berlebih, dan kesalahan penulisan ("Cloths", "Electronic", "Mid-West"). Standardisasi dilakukan melalui kolom bantu dengan formula PROPER + TRIM, dilanjutkan koreksi typo manual via Find & Replace, lalu hasilnya di-paste kembali (Paste Special > Values) ke kolom asli:

```
=PROPER(TRIM([@category]))
```

|    | A         | B          | C           | D              | E        | F           | G                 | H        | I          | J            | K              | L                 | M |
|----|-----------|------------|-------------|----------------|----------|-------------|-------------------|----------|------------|--------------|----------------|-------------------|---|
|    | order_id  | order_date | customer_id | customer_name  | region   | category    | product_name      | quantity | unit_price | order_status | payment_method | category_clean    |   |
| 1  | ORD100016 | 11/23/2023 | CUST1353    | Andi Halim     |          | Sports & Ou | Water Bottle      | 2        | 28.97      | Completed    | Credit Card    | Sports & Outdoors |   |
| 2  | ORD102704 | 3/14/2023  | CUST1478    | Lisa Kurniawan | East     | Toys        | Building Blocks   | 2        | 46.3       | Returned     | Credit Card    | F3))              |   |
| 4  | ORD103490 | 4/10/2024  | CUST1531    | Ahmad Halim    | Mid-West | Beauty      | Moisturizer       | 1        | 35.09      | Completed    | Gift Card      | Beauty            |   |
| 5  | ORD101567 | 8/22/2024  | CUST1390    | Ayu Hidayat    | SOUTH    | Sports & Ou | Camping Tent      | 2        | 12.94      | Completed    | Debit Card     | Sports & Outdoors |   |
| 6  | ORD101096 | 21-Feb-24  | CUST1563    | James Yusuf    | SOUTH    | sports & ou | Hiking Backpack   | 1        | 23.12      | Completed    | PayPal         | Sports & Outdoors |   |
| 7  | ORD101473 | 8/18/2024  | CUST1052    | Fajar Halim    | Midwest  | Beauty      | Sunscreen SPF50   | 1        | 33.88      | Completed    | Gift Card      | Beauty            |   |
| 8  | ORD103111 | 8/15/2024  | CUST1726    | John Tan       | south    | books       | Fiction Novel     | 2        | 10.88      | Completed    | Gift Card      | Books             |   |
| 9  | ORD100537 | 1/28/2023  | CUST1445    | Kevin Anderson | WEST     | Beauty      | Moisturizer       | 1        | 21.46      | Completed    | PayPal         | Beauty            |   |
| 10 | ORD102413 | 24-Apr-23  | CUST1699    | John Yusuf     | MIDWEST  | Home & Kit  | Cutting Board     | 1        | 102.66     | Completed    | PayPal         | Home & Kitchen    |   |
| 11 | ORD103151 | 11-Aug-23  | CUST1708    | Ayu Smith      | South    | sports & ou | Resistance Bands  | 1        | 56.4       | Completed    | PayPal         | Sports & Outdoors |   |
| 12 | ORD102767 | 9/23/2023  | CUST1868    | Nina Lee       | midwest  | Books       | Business Strategy | 1        | 18.14      | Completed    | Credit Card    | Books             |   |
| 13 | ORD101868 | 15-Mar-24  | CUST1211    |                | south    | Clothing    | Winter Jacket     | 2        | 35.68      | Completed    | Gift Card      | Clothing          |   |
| 14 | ORD100749 | 7/27/2024  | CUST1334    | Fajar Santos   | east     | Home & Kit  | Storage Container | 1        | 112.68     | Completed    | Gift Card      | Home & Kitchen    |   |
| 15 | ORD101857 | 8/5/2023   | CUST1104    | Sarah Tan      | East     | Electronics | Webcam            | 1        | 68.92      | Completed    | Debit Card     | Electronics       |   |
| 16 | ORD100800 | 23-Jan-24  | CUST1352    | Lisa Taylor    | MIDWEST  | Beauty      | Shampoo           | 2        | 23.18      | Cancelled    |                | Beauty            |   |
| 17 | ORD101695 | 20-May-24  | CUST1255    | Budi Kurniawan | West     | Clothing    | Hoodie            | 2        | 66.21      | Returned     | PayPal         | Clothing          |   |
| 18 | ORD102093 | 1/25/2024  | CUST1824    | Linda Chen     | east     | Sports & Ou | Camping Tent      | 1        |            | Cancelled    | PayPal         | Sports & Outdoors |   |
| 19 | ORD101296 | 10/6/2024  | CUST1003    | Siti Anderson  | East     | Electronics | Power Bank        | 3        | 134.85     | Completed    | PayPal         | Electronics       |   |
| 20 | ORD103842 | 9/17/2023  | CUST1741    | Siti Wilson    | midwest  | Sports & Ou | Yoga Mat          | 1        | 75.14      | Completed    | Credit Card    | Sports & Outdoors |   |
| 21 | ORD102931 | 10/26/2023 | CUST1856    |                | SOUTH    | electronics | Power Bank        | 5        | 126.65     | Completed    | Gift Card      | Electronics       |   |
| 22 | ORD102954 | 8/19/2023  | CUST1210    | Nina Garcia    | West     | Sports & Ou | Hiking Backpack   | 1        | 49.73      | Returned     | Credit Card    | Sports & Outdoors |   |
| 23 | ORD103824 | 7/22/2024  | CUST1524    | Robert Smith   | west     | clothing    | Denim Jeans       | 4        | 62.83      | Completed    | Debit Card     | Clothing          |   |
| 24 | ORD101074 | 4/12/2024  | CUST1689    | Siti Tan       | EAST     | Beauty      | Face Mask Set     | 1        | 39.05      | Completed    | Debit Card     | Beauty            |   |

Gambar 4. Kolom bantu category\_clean memakai formula PROPER(TRIM()))

|    | A         | B          | C           | D              | E       | F                 | G                 | H        | I          | J            | K              | L           | M |
|----|-----------|------------|-------------|----------------|---------|-------------------|-------------------|----------|------------|--------------|----------------|-------------|---|
|    | order_id  | order_date | customer_id | customer_name  | region  | category          | product_name      | quantity | unit_price | order_status | payment_method | region_trim |   |
| 1  | ORD100016 | 2023-11-23 | CUST1353    | Andi Halim     |         | Sports & Outdoors | Water Bottle      | 2        | 28.97      | Completed    | Credit Card    |             |   |
| 2  | ORD102704 | 2023-03-14 | CUST1478    | Lisa Kurniawan | East    | Toys              | Building Blocks   | 2        | 46.3       | Returned     | Credit Card    | East        |   |
| 3  | ORD103490 | 2024-04-10 | CUST1531    | Ahmad Halim    | Midwest | Beauty            | Moisturizer       | 1        | 35.09      | Completed    | Gift Card      | Midwest     |   |
| 4  | ORD101567 | 2024-08-22 | CUST1390    | Ayu Hidayat    | South   | Sports & Outdoors | Camping Tent      | 2        | 12.94      | Completed    | Debit Card     | South       |   |
| 5  | ORD101096 | 2024-02-21 | CUST1563    | James Yusuf    | South   | Sports & Outdoors | Hiking Backpack   | 1        | 23.12      | Completed    | PayPal         | South       |   |
| 6  | ORD101473 | 2024-08-18 | CUST1052    | Fajar Halim    | Midwest | Beauty            | Sunscreen SPF50   | 1        | 33.88      | Completed    | Gift Card      | Midwest     |   |
| 7  | ORD103111 | 2024-08-15 | CUST1726    | John Tan       | South   | Books             | Fiction Novel     | 2        | 10.88      | Completed    | Gift Card      | South       |   |
| 8  | ORD100537 | 2024-01-28 | CUST1445    | Kevin Anderson | West    | Beauty            | Moisturizer       | 1        | 21.46      | Completed    | PayPal         | West        |   |
| 9  | ORD102413 | 2023-04-24 | CUST1699    | John Yusuf     | Midwest | Home & Kitchen    | Cutting Board     | 1        | 102.66     | Completed    | PayPal         | Midwest     |   |
| 10 | ORD103151 | 2023-08-11 | CUST1708    | Ayu Smith      | South   | Sports & Outdoors | Resistance Bands  | 1        | 56.4       | Completed    | PayPal         | South       |   |
| 11 | ORD102767 | 2023-09-22 | CUST1868    | Nina Lee       | Midwest | Books             | Business Strategy | 1        | 18.14      | Completed    | Credit Card    | Midwest     |   |
| 12 | ORD101868 | 2024-03-15 | CUST1211    |                | South   | Clothing          | Winter Jacket     | 2        | 35.68      | Completed    | Gift Card      | South       |   |
| 13 | ORD100749 | 2024-07-27 | CUST1334    | Fajar Santos   | East    | Home & Kitchen    | Storage Container | 1        | 112.68     | Completed    | Gift Card      | East        |   |
| 14 | ORD101857 | 2023-08-05 | CUST1104    | Sarah Tan      | East    | Electronics       | Webcam            | 1        | 68.92      | Completed    | Debit Card     | East        |   |
| 15 | ORD100800 | 2024-02-23 | CUST1352    | Lisa Taylor    | Midwest | Beauty            | Shampoo           | 2        | 23.18      | Cancelled    |                | Midwest     |   |
| 16 | ORD101695 | 2024-05-20 | CUST1255    | Budi Kurniawan | West    | Clothing          | Hoodie            | 2        | 66.21      | Returned     | PayPal         | West        |   |
| 17 | ORD102093 | 2024-01-25 | CUST1824    | Linda Chen     | East    | Sports & Outdoors | Camping Tent      | 1        |            | Cancelled    | PayPal         | East        |   |
| 18 | ORD101296 | 2024-10-06 | CUST1003    | Siti Anderson  | East    | Electronics       | Power Bank        | 3        | 134.85     | Completed    | PayPal         | East        |   |
| 19 | ORD103842 | 2023-09-17 | CUST1741    | Siti Wilson    | Midwest | Sports & Outdoors | Yoga Mat          | 1        | 75.14      | Completed    | Credit Card    | Midwest     |   |
| 20 | ORD102931 | 2023-10-26 | CUST1856    |                | South   | Electronics       | Power Bank        | 5        | 126.65     | Completed    | Gift Card      | South       |   |
| 21 | ORD102954 | 2023-08-19 | CUST1210    | Nina Garcia    | West    | Sports & Outdoors | Hiking Backpack   | 1        | 49.73      | Returned     | Credit Card    | West        |   |
| 22 | ORD103824 | 2024-07-22 | CUST1524    | Robert Smith   | West    | Clothing          | Denim Jeans       | 4        | 62.83      | Completed    | Debit Card     | West        |   |
| 23 | ORD101074 | 2024-04-12 | CUST1689    | Siti Tan       | East    | Beauty            | Face Mask Set     | 1        | 39.05      | Completed    | Debit Card     | East        |   |
| 24 | ORD101777 | 2024-12-28 | CUST1227    | David Wijaya   | East    | Home & Kitchen    | Blender           | 4        | 115.54     | Completed    | Debit Card     | East        |   |
| 25 | ORD102224 | 2023-05-05 | CUST1588    | David Anderson | South   | Sports & Outdoors | Camping Tent      | 1        | 30.3       | Completed    | Debit Card     | South       |   |
| 26 | ORD103815 | 2024-07-04 | CUST1265    | Sarah Johnson  | South   | Clothing          | Hoodie            | 1        | 46.19      | Completed    | Debit Card     | South       |   |
| 27 | ORD101819 | 2023-02-09 | CUST1343    | Dewi Sari      | East    | Books             | Self-Help Guide   | 1        | 25.66      | Completed    | Credit Card    | East        |   |
| 28 | ORD100756 | 2023-07-17 | CUST1680    | Putri Halim    | East    | Sports & Outdoors | Camping Tent      | 1        | 72.43      | Cancelled    | PayPal         | East        |   |
| 29 | ORD101910 | 2023-04-08 | CUST1987    | Maria Wilton   | South   | Toys              | Board Game        | 1        | 58.15      | Completed    | PayPal         | South       |   |
| 30 | ORD100326 | 2024-07-07 | CUST1182    | Dewi Tan       | East    | Beauty            | Lip Balm          | 2        | 7.83       | Completed    | PayPal         | East        |   |
| 31 | ORD103238 | 2023-04-16 | CUST1254    | Nina Garcia    | Midwest | Toys              | RC Car            | 1        | 9.57       | Completed    |                | Midwest     |   |
| 32 | ORD102634 | 2023-05-24 | CUST1757    | Sarah Lee      | East    | Clothing          | Cotton T-Shirt    | 1        | 54.55      | Completed    | Debit Card     | East        |   |
| 33 | ORD103616 | 2024-02-26 | CUST1128    | Rina Taylor    | South   | Beauty            | Moisturizer       | 1        | 25.47      | Completed    | Debit Card     | South       |   |
| 34 | ORD101939 | 2023-11-15 | CUST1806    | Ahmad Hidayat  | Midwest | Home & Kitchen    | Coffee Maker      | 2        | 52.95      | Cancelled    | Gift Card      | Midwest     |   |
| 35 | ORD103226 | 2024-09-07 | CUST1206    | Siti Wilson    | West    | Beauty            | Shampoo           | 2        | \$44.80    | Completed    | PayPal         | West        |   |
| 36 | ORD102507 | 2023-09-23 | CUST1645    | Sarah Chen     | West    | Toys              | Board Game        | 1        | 58.26      | Completed    | Debit Card     | West        |   |
| 37 | ORD100857 | 2023-01-17 | CUST1370    | Rina Anderson  | West    | Toys              | Puzzle 1000pc     | 1        | 25.68      | Completed    | Debit Card     | West        |   |
| 38 | ORD102423 | 2023-05-01 | CUST1838    | Budi Chen      | South   | Electronics       | Bluetooth Speaker | 2        | 93.44      | Completed    | PayPal         | South       |   |
| 39 | ORD100333 | 2024-09-25 | CUST1753    | Kevin Lee      | East    | Clothing          | Running Shorts    | 2        | 24.84      | Completed    | PayPal         | East        |   |
| 40 | ORD102984 | 2023-09-17 | CUST1615    | Putri Anderson | West    | Clothing          | Hoodie            | 2        |            | Completed    | PayPal         | West        |   |
| 41 | ORD100087 | 2023-03-06 | CUST1867    | Nina Garcia    | South   | Beauty            | Moisturizer       | 2        | 10.48      | Completed    | Gift Card      | South       |   |
| 42 | ORD100619 | 2024-06-21 | CUST1414    | Ayu Chen       | South   | Electronics       | Power Bank        | 2        | 142.03     | Completed    | Debit Card     | South       |   |

Gambar 5. Kolom category dan region setelah distandarisasi (Proper Case, tanpa typo)

### 3.3 Standardisasi Format Tanggal (order\_date)

Kolom order\_date awalnya berisi 5 format berbeda (YYYY-MM-DD, MM/DD/YYYY, DD-Mon-YYYY, DD/MM/YYYY, YYYY/MM/DD). Baris yang masih tersimpan sebagai teks diidentifikasi terlebih dahulu dengan ISTEXT:

```
=ISTEXT([@order_date])
```

|  | id | is_text_date |
|--|----|--------------|
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
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|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
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|  |    | FALSE        |
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|  |    | FALSE        |
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|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | FALSE        |
|  |    | TRUE         |
|  |    | FALSE        |
|  |    | FALSE        |

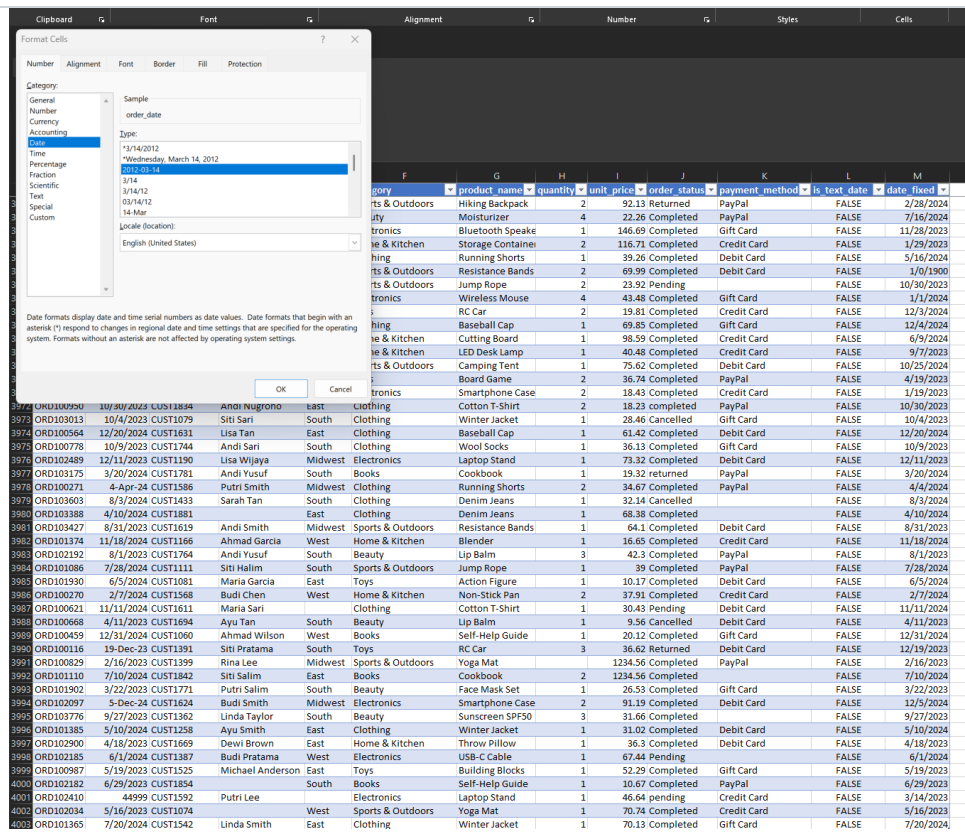
Gambar 6. Kolom bantu `is_text_date — TRUE` menandakan baris yang `order_date`-nya masih tersimpan sebagai teks

Baris yang bernilai TRUE dikonversi menggunakan formula berikut, yang membaca format DD/MM/YYYY secara eksplisit dan membiarkan tanggal yang sudah valid (FALSE) tidak berubah:

```
=IF(ISTEXT([@order_date]), DATE(VALUE(RIGHT([@order_date],4)),  
VALUE(MID([@order_date],4,2)), VALUE(LEFT([@order_date],2))), [@order_date])
```

Setelah seluruh nilai berhasil dikonversi menjadi tanggal asli, format tampilan diseragamkan lewat Format Cells > Date > custom type YYYY-MM-DD (bukan format beraster yang mengikuti regional setting Windows):





Gambar 7. Dialog Format Cells — memilih format tanggal YYYY-MM-DD yang konsisten di semua sistem

|    | A         | B          | C           | D              | E       | F                 | G                  | H        | I          | J            | K              | L |
|----|-----------|------------|-------------|----------------|---------|-------------------|--------------------|----------|------------|--------------|----------------|---|
|    | order_id  | order_date | customer_id | customer_name  | region  | category          | product_name       | quantity | unit_price | order_status | payment_method |   |
| 1  | ORD100016 | 2023-11-23 | CUST1353    | Andi Halim     | East    | Sports & Outdoors | Water Bottle       | 2        | 28.97      | Completed    | Credit Card    |   |
| 2  | ORD102704 | 2023-03-14 | CUST1478    | Lisa Kurniawan | East    | Toys              | Building Blocks    | 2        | 46.3       | Returned     | Credit Card    |   |
| 4  | ORD103490 | 2024-04-10 | CUST1531    | Ahmad Halim    | Midwest | Beauty            | Moisturizer        | 1        | 35.09      | Completed    | Gift Card      |   |
| 5  | ORD101567 | 2024-08-22 | CUST1390    | Ayu Hidayat    | SOUTH   | Sports & Outdoors | Camping Tent       | 2        | 12.94      | Completed    | Debit Card     |   |
| 6  | ORD101096 | 2024-02-21 | CUST1563    | James Yusuf    | SOUTH   | Sports & Outdoors | Hiking Backpack    | 1        | 23.12      | Completed    | PayPal         |   |
| 7  | ORD101473 | 2024-08-18 | CUST1052    | Fajar Halim    | Midwest | Beauty            | Sunscreens SPF50   | 1        | 33.88      | Completed    | Gift Card      |   |
| 8  | ORD103111 | 2024-08-15 | CUST1726    | John Tan       | south   | Books             | Fiction Novel      | 2        | 10.88      | Completed    | Gift Card      |   |
| 9  | ORD100537 | 2024-01-28 | CUST1445    | Kevin Anderson | WEST    | Beauty            | Moisturizer        | 1        | 21.46      | Completed    | PayPal         |   |
| 10 | ORD102413 | 2023-04-24 | CUST1699    | John Yusuf     | MIDWEST | Home & Kitchen    | Cutting Board      | 1        | 102.66     | Completed    | PayPal         |   |
| 11 | ORD103151 | 2023-08-11 | CUST1708    | Ayu Smith      | South   | Sports & Outdoors | Resistance Bands   | 1        | 56.4       | Completed    | PayPal         |   |
| 12 | ORD102767 | 2023-09-22 | CUST1868    | Nina Lee       | midwest | Books             | Business Strategy  | 1        | 18.14      | Completed    | Credit Card    |   |
| 13 | ORD101868 | 2024-03-15 | CUST1211    |                | south   | Clothing          | Winter Jacket      | 2        | 35.68      | Completed    | Gift Card      |   |
| 14 | ORD100749 | 2024-07-27 | CUST1334    | Fajar Santos   | east    | Home & Kitchen    | Storage Containers | 1        | 112.68     | Completed    | Gift Card      |   |
| 15 | ORD101857 | 2023-08-05 | CUST1104    | Sarah Tan      | East    | Electronics       | Webcam             | 1        | 68.92      | Completed    | Debit Card     |   |
| 16 | ORD100800 | 2024-01-23 | CUST1352    | Lisa Taylor    | MIDWEST | Beauty            | Shampoo            | 2        | 23.18      | Cancelled    |                |   |
| 17 | ORD101695 | 2024-05-20 | CUST1255    | Budi Kurniawan | West    | Clothing          | Hoodie             | 2        | 66.21      | Returned     | PayPal         |   |
| 18 | ORD102093 | 2024-01-25 | CUST1824    | Linda Chen     | east    | Sports & Outdoors | Camping Tent       | 1        |            | Cancelled    | PayPal         |   |
| 19 | ORD101296 | 2024-10-06 | CUST1003    | Siti Anderson  | East    | Electronics       | Power Bank         | 3        | 134.85     | Completed    | PayPal         |   |
| 20 | ORD103842 | 2023-09-17 | CUST1741    | Siti Wilson    | midwest | Sports & Outdoors | Yoga Mat           | 1        | 75.14      | Completed    | Credit Card    |   |
| 21 | ORD102931 | 2023-10-26 | CUST1856    |                | SOUTH   | Electronics       | Power Bank         | 5        | 126.65     | Completed    | Gift Card      |   |
| 22 | ORD102954 | 2023-08-19 | CUST1210    | Nina Garcia    | West    | Sports & Outdoors | Hiking Backpack    | 1        | 49.73      | Returned     | Credit Card    |   |
| 23 | ORD103824 | 2024-07-22 | CUST1524    | Robert Smith   | west    | Clothing          | Denim Jeans        | 4        | 62.83      | Completed    | Debit Card     |   |
| 24 | ORD101074 | 2024-04-12 | CUST1689    | Siti Tan       | EAST    | Beauty            | Face Mask Set      | 1        | 39.05      | Completed    | Debit Card     |   |
| 25 | ORD101777 | 2024-12-28 | CUST1227    | David Wijaya   | EAST    | Home & Kitchen    | Blender            | 4        | 115.54     | Completed    | Debit Card     |   |
| 26 | ORD102224 | 2023-05-05 | CUST1588    | David Anderson | South   | Sports & Outdoors | Camping Tent       | 1        | 30.3       | Completed    | Debit Card     |   |
| 27 | ORD103815 | 2024-07-04 | CUST1265    | Sarah Johnson  | south   | Clothing          | Hoodie             | 1        | 46.19      | Completed    | Debit Card     |   |
| 28 | ORD101819 | 2023-02-09 | CUST1343    | Dewi Sari      | EAST    | Books             | Self-Help Guide    | 1        | 25.66      | Completed    | Credit Card    |   |
| 29 | ORD100756 | 2023-07-17 | CUST1680    | Putri Halim    | East    | Sports & Outdoors | Camping Tent       | 1        | 72.43      | Cancelled    | PayPal         |   |
| 30 | ORD101910 | 2023-04-08 | CUST1887    | Maria Wilson   | SOUTH   | Toys              | Board Game         | 1        | 58.15      | Completed    | PayPal         |   |
| 31 | ORD100326 | 2024-07-07 | CUST1182    | Dewi Tan       | east    | Beauty            | Lip Balm           | 2        | 7.83       | Completed    | PayPal         |   |
| 32 | ORD103238 | 2023-04-16 | CUST1254    | Nina Garcia    | MIDWEST | Toys              | RC Car             | 1        | 9.57       | Completed    |                |   |
| 33 | ORD102634 | 2023-05-24 | CUST1757    | Sarah Lee      | East    | Clothing          | Cotton T-Shirt     | 1        | 54.55      | Completed    | Debit Card     |   |
| 34 | ORD103616 | 2024-02-26 | CUST1128    | Rina Taylor    | SOUTH   | Beauty            | Moisturizer        | 1        | 25.47      | Completed    | Debit Card     |   |
| 35 | ORD101939 | 2023-11-15 | CUST1806    | Ahmad Hidayat  | Midwest | Home & Kitchen    | Coffee Maker       | 2        | 52.95      | Cancelled    | Gift Card      |   |
| 36 | ORD103226 | 2024-09-07 | CUST1206    | Siti Wilson    | WEST    | Beauty            | Shampoo            | 2        | \$44.80    | Completed    | PayPal         |   |
| 37 | ORD102507 | 2023-09-23 | CUST1645    | Sarah Chen     | WEST    | Toys              | Board Game         | 1        | 58.26      | Completed    | Debit Card     |   |
| 38 | ORD100857 | 2023-01-17 | CUST1370    | Rina Anderson  | WEST    | Toys              | Puzzle 1000pc      | 1        | 25.68      | Completed    | Debit Card     |   |
| 39 | ORD102423 | 2023-05-01 | CUST1838    | Budi Chen      | South   | Electronics       | Bluetooth Speaker  | 2        | 93.44      | Completed    | PayPal         |   |
| 40 | ORD100333 | 2024-09-25 | CUST1753    | Kevin Lee      | East    | Clothing          | Running Shorts     | 2        | 24.84      | Completed    | PayPal         |   |
| 41 | ORD102984 | 2023-09-17 | CUST1615    | Putri Anderson | West    | Clothing          | Hoodie             | 2        |            | Completed    | PayPal         |   |
| 42 | ORD100087 | 2023-03-06 | CUST1867    | Nina Garcia    | SOUTH   | Beauty            | Moisturizer        | 2        | 10.48      | Completed    | Gift Card      |   |
| 43 | ORD100619 | 2024-06-21 | CUST1414    | Ayu Chen       | South   | Electronics       | Power Bank         | 2        | 142.03     | Completed    | Debit Card     |   |

Gambar 8. Kolom order\_date setelah seluruh nilai dikonversi dan diformat seragam

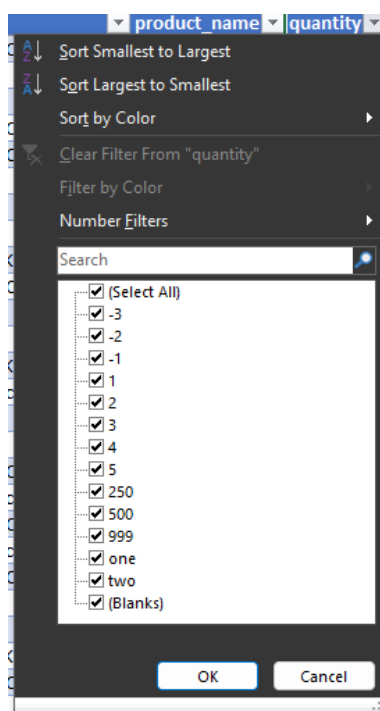
- Seluruh kolom order\_date diformat ulang secara seragam ke YYYY-MM-DD.
- Satu nilai tanggal yang tersimpan sebagai 0 (bekas sel kosong) ditemukan lewat pengecekan `=MIN(order_date)` dan dikembalikan menjadi blank agar tidak menyamarkan missing value sebagai tanggal 1900.

|    | A     | B                   | C                                               | D     | E        | F |
|----|-------|---------------------|-------------------------------------------------|-------|----------|---|
| 1  | Hari  | Fase                | Task                                            | Tools | Selesai? |   |
| 2  | Day 1 | Excel Data Cleaning | Ambil dataset e-commerce sales                  | Excel | SELESAI  |   |
| 3  | Day 1 | Excel Data Cleaning | Cek & tangani missing values                    | Excel |          |   |
| 4  | Day 1 | Excel Data Cleaning | Cek & hapus duplicate records                   | Excel | SELESAI  |   |
| 5  | Day 1 | Excel Data Cleaning | Perbaiki data types yang salah                  | Excel |          |   |
| 6  | Day 1 | Excel Data Cleaning | Standarkan kategori yang tidak konsisten        | Excel | SELESAI  |   |
| 7  | Day 1 | Excel Data Cleaning | Format ulang kolom tanggal                      | Excel | SELESAI  |   |
| 8  | Day 1 | Excel Data Cleaning | Cek outlier                                     | Excel |          |   |
| 9  | Day 1 | Excel Data Cleaning | Buat calculated columns                         | Excel |          |   |
| 10 | Day 1 | Excel Data Cleaning | Dokumentasikan raw -> cleaning -> clean dataset | Excel |          |   |
| 11 | Day 1 | Excel Data Cleaning | Simpan sebagai cleaned sales data.xlsx          | Excel |          |   |

Gambar 9. Checklist Day 1 setelah tahap duplicate, data types, category/region, dan tanggal selesai

### 3.4 Perbaiki Tipe Data dan Outlier (Quantity & Unit Price)

Kolom quantity dan unit\_price memiliki beberapa masalah sekaligus: nilai bertipe teks ("one", "two"), nilai negatif, harga tersimpan sebagai teks berformat "\$xx.xx", serta outlier ekstrem yang tidak realistis untuk transaksi retail individual.



Gambar 10. Variasi nilai bermasalah pada kolom quantity, dilihat lewat filter dropdown (negatif, teks, dan outlier ekstrem tercampur)

| G             | H        | I          | J            | K              | L              |
|---------------|----------|------------|--------------|----------------|----------------|
| product_name  | quantity | unit_price | order_status | payment_method | quantity_clean |
| Wool Socks    | -3       | 64.63      | Completed    | PayPal         | 3              |
| Fiction Novel | -3       | 30.39      | Cancelled    | Credit Card    | 3              |

Gambar 11. Contoh baris quantity bernilai negatif (-3) sebelum diperbaiki

#### Perbaiki tipe data quantity

```
quantity_clean:
=IFERROR(IF([@quantity]="one",1,IF([@quantity]="two",2,IF([@quantity]="three",3,ABS(VALUE([@quantity]))))),"
```

#### Perbaiki tipe data unit\_price

|  | H        | I          | J            | K              | L              |
|--|----------|------------|--------------|----------------|----------------|
|  | quantity | unit_price | order_status | payment_method | quantity_clean |
|  | 2        | \$44.80    | Completed    | PayPal         | 2              |
|  | 1        | \$75.09    | Completed    | PayPal         | 1              |
|  | 1        | \$86.87    | Completed    | PayPal         | 1              |
|  | 1        | \$69.18    | Completed    | Credit Card    | 1              |
|  | 1        | \$17.17    | Completed    | Debit Card     | 1              |
|  | 1        | \$60.06    | Completed    | Gift Card      | 1              |
|  | 1        | \$20.53    | Completed    | PayPal         | 1              |
|  | 1        | \$31.32    | Completed    | PayPal         | 1              |
|  | 1        | \$106.64   | Completed    | Debit Card     | 1              |
|  | 2        | \$34.82    | Completed    | Credit Card    | 2              |
|  | 1        | \$41.30    | Completed    | Gift Card      | 1              |
|  | 1        | \$130.75   | Completed    | Credit Card    | 1              |
|  | 1        | \$29.64    | Completed    |                | 1              |
|  | 1        | \$16.59    | completed    | Debit Card     | 1              |
|  | 1        | \$38.05    | Completed    | Gift Card      | 1              |
|  | 1        | \$44.88    | Completed    | Credit Card    | 1              |
|  | 2        | \$12.32    | Completed    | Debit Card     | 2              |
|  | 2        | \$75.42    | Completed    | Debit Card     | 2              |
|  | 3        | \$15.48    | cancelled    | Gift Card      | 3              |

Gambar 12. Kolom unit\_price masih berformat teks "\$xx.xx" sebelum dibersihkan

```
unit_price_clean:
=IFERROR(VALUE(SUBSTITUTE([@unit_price],"$","")), "")
```

|  | H        | I          | J            | K              | L              | M          |
|--|----------|------------|--------------|----------------|----------------|------------|
|  | quantity | unit_price | order_status | payment_method | quantity_clean | unit_price |
|  | 2        | \$44.80    | Completed    | PayPal         | 2              | 44.8       |
|  | 1        | \$75.09    | Completed    | PayPal         | 1              | 75.09      |
|  | 1        | \$86.87    | Completed    | PayPal         | 1              | 86.87      |
|  | 1        | \$69.18    | Completed    | Credit Card    | 1              | 69.18      |
|  | 1        | \$17.17    | Completed    | Debit Card     | 1              | 17.17      |
|  | 1        | \$60.06    | Completed    | Gift Card      | 1              | 60.06      |
|  | 1        | \$20.53    | Completed    | PayPal         | 1              | 20.53      |
|  | 1        | \$31.32    | Completed    | PayPal         | 1              | 31.32      |
|  | 1        | \$106.64   | Completed    | Debit Card     | 1              | 106.64     |
|  | 2        | \$34.82    | Completed    | Credit Card    | 2              | 34.82      |
|  | 1        | \$41.30    | Completed    | Gift Card      | 1              | 41.3       |
|  | 1        | \$130.75   | Completed    | Credit Card    | 1              | 130.75     |
|  | 1        | \$29.64    | Completed    |                | 1              | 29.64      |
|  | 1        | \$16.59    | completed    | Debit Card     | 1              | 16.59      |
|  | 1        | \$38.05    | Completed    | Gift Card      | 1              | 38.05      |
|  | 1        | \$44.88    | Completed    | Credit Card    | 1              | 44.88      |
|  | 2        | \$12.32    | Completed    | Debit Card     | 2              | 12.32      |
|  | 2        | \$75.42    | Completed    | Debit Card     | 2              | 75.42      |
|  | 3        | \$15.48    | cancelled    | Gift Card      | 3              | 15.48      |

Gambar 13. Kolom unit\_price\_clean — tanda \$ berhasil dihapus, tersisa angka murni

## Identifikasi dan penghapusan outlier

Setelah tipe data bersih, outlier diperiksa dengan mengurutkan (Sort Largest to Smallest) dan meninjau konteks tiap baris (produk, customer, kategori):

|    | A         | B          | C           | D                 | E       | F                 | G                 | H        | I          | J            | K              | L              | M          | N |
|----|-----------|------------|-------------|-------------------|---------|-------------------|-------------------|----------|------------|--------------|----------------|----------------|------------|---|
|    | order_id  | order_date | customer_id | customer_name     | region  | category          | product_name      | quantity | unit_price | order_status | payment_method | quantity_clean | unit_price |   |
| 1  | ORD100335 | 2023-08-07 | CUST11411   | Ahmad Garcia      | West    | Home & Kitchen    | Coffee Maker      | 999      | 43.04      | Completed    | PayPal         | 999            | 43.04      |   |
| 2  | ORD101178 | 2023-08-01 | CUST11004   | Ahmad Yusuf       | South   | Toys              | Action Figure     | 999      | 35.43      | completed    | Gift Card      | 999            | 35.43      |   |
| 3  | ORD102638 | 2023-02-23 | CUST11083   | Ahmad Taylor      | East    | Sports & Outdoors | Yoga Mat          | 999      | 71.18      | Completed    | Gift Card      | 999            | 71.18      |   |
| 4  | ORD103218 | 2023-10-28 | CUST11474   | Kevin Sari        | East    | Beauty            | Sunscreen SPF50   | 999      | 9.1        | Completed    | Gift Card      | 999            | 9.1        |   |
| 5  | ORD102875 | 2023-04-30 | CUST11808   | Michael Kurniawan | Midwest | Sports & Outdoors | Jump Rope         | 999      | 85.82      | Completed    | Credit Card    | 999            | 85.82      |   |
| 6  | ORD100684 | 2023-09-16 | CUST11652   | Ahmad Smith       | West    | Books             | Business Strategy | 999      | 17.71      | Completed    | Debit Card     | 999            | 17.71      |   |
| 7  | ORD102343 | 2023-07-25 | CUST11772   | Ahmad Hidayat     |         | Electronics       | Wireless Earbuds  | 500      | 52.43      | Returned     | Gift Card      | 500            | 52.43      |   |
| 8  | ORD103588 | 2023-01-01 | CUST11104   | Sarah Tan         | South   | Sports & Outdoors | Water Bottle      | 500      | 32.5       | Cancelled    | PayPal         | 500            | 32.5       |   |
| 9  | ORD100867 | 2023-01-31 | CUST11399   | Dewi Chen         | West    | Clothing          | Winter Jacket     | 500      | 32.44      | Completed    | Gift Card      | 500            | 32.44      |   |
| 10 | ORD102002 | 2023-05-27 | CUST11262   | Robert Garcia     | Midwest | Clothing          | Hoodie            | 500      | 13.13      | Completed    | Credit Card    | 500            | 13.13      |   |
| 11 | ORD101689 | 2023-07-10 | CUST11290   | Lisa Nugroho      | West    | Home & Kitchen    | Non-Stick Pan     | 500      | 98.01      | Completed    | Debit Card     | 500            | 98.01      |   |
| 12 | ORD100031 | 2024-05-22 | CUST11179   | David Salim       | East    | Sports & Outdoors | Water Bottle      | 250      | 9.42       | Completed    | Debit Card     | 250            | 9.42       |   |
| 13 | ORD100209 | 2024-01-20 | CUST11761   | Ahmad Johnson     | East    | Toys              | Building Blocks   | 250      | 53.78      | Completed    |                | 250            | 53.78      |   |
| 14 | ORD103734 | 2024-12-01 | CUST11871   | Andi Smith        | West    | Electronics       | Laptop Stand      | 250      | 127.77     | Cancelled    | Debit Card     | 250            | 127.77     |   |
| 15 | ORD102727 | 2023-09-04 | CUST11713   | Dewi Taylor       | West    | Electronics       | Webcam            | 250      | 44.7       | Completed    | Debit Card     | 250            | 44.7       |   |
| 16 | ORD102931 | 2023-10-26 | CUST11856   |                   | South   | Electronics       | Power Bank        | 5        | 126.65     | Completed    | Gift Card      | 5              | 126.65     |   |
| 17 | ORD103343 | 2023-12-09 | CUST11383   | David Garcia      | West    | Sports & Outdoors | Hiking Backpack   | 5        | 68.67      | Completed    | Debit Card     | 5              | 68.67      |   |
| 18 | ORD100234 | 2023-12-22 | CUST11766   | Michael Kurniawan | South   | Beauty            | Sunscreen SPF50   | 5        | 37.89      | Completed    | PayPal         | 5              | 37.89      |   |
| 19 | ORD102815 | 2023-01-17 | CUST11555   | Andi Hidayat      | Midwest | Toys              | Action Figure     | 5        | 48.39      | Completed    | Debit Card     | 5              | 48.39      |   |

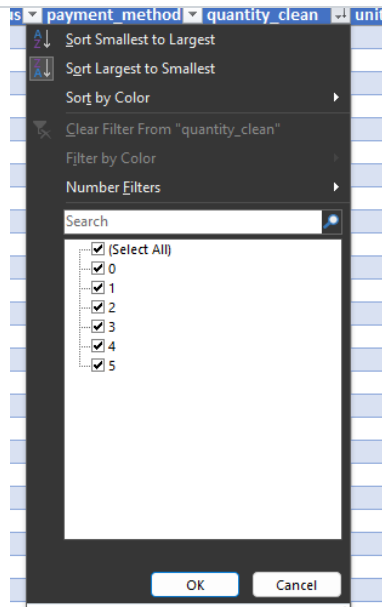
Gambar 14. Kolom quantity\_clean diurutkan dari terbesar — terlihat 15 baris dengan nilai 999/500/250

|    |           |            |           |             |         |      |               |   |      |           |            |   |      |  |
|----|-----------|------------|-----------|-------------|---------|------|---------------|---|------|-----------|------------|---|------|--|
| 19 | ORD100234 | 2024-09-25 | CUST11716 | Budi Salim  | Midwest | Toys | Puzzle 1000pc | 5 | 8888 | Completed | Gift Card  | 5 | 8888 |  |
| 20 | ORD101663 | 2024-08-01 | CUST11572 | Maria Yusuf | East    | Toys | Puzzle 1000pc | 5 | 51.6 | Cancelled | Debit Card | 5 | 51.6 |  |

Gambar 15. Salah satu baris outlier quantity (Puzzle 1000pc) kebetulan juga memiliki unit\_price ekstrem (8888)

- Quantity: 15 baris dengan nilai 250/500/999 dihapus. Nilai tersebar acak di berbagai produk dan customer tanpa pola order borongan (wholesale) — diklasifikasikan sebagai data entry error, dan immaterial karena hanya 0,37% dari dataset.





Gambar 16. Filter kolom quantity setelah 15 baris outlier dihapus — tersisa nilai 0–5 saja

|    |           |            |          |                |         |                   |                   |   |                   |             |   |         |
|----|-----------|------------|----------|----------------|---------|-------------------|-------------------|---|-------------------|-------------|---|---------|
| 77 | ORD101914 | 2023-09-23 | CUST1838 | Budi Chen      | South   | Home & Kitchen    | Throw Pillow      |   | Completed         | PayPal      |   | 0       |
| 78 | ORD100143 | 2024-10-24 | CUST1799 | Robert Hidayat | South   | Toys              | Building Blocks   |   | Completed         | Credit Card |   | 0       |
| 79 | ORD100533 | 2024-09-25 | CUST1716 | Budi Salim     | Midwest | Toys              | Puzzle 1000pc     | 5 | 8888 Completed    | Gift Card   | 5 | 8888    |
| 80 | ORD100693 | 2024-03-23 | CUST1630 | John Kurniawan | West    | Home & Kitchen    | Blender           | 1 | 8888 Returned     | PayPal      | 1 | 8888    |
| 81 | ORD101843 | 2024-12-16 | CUST1023 | Siti Wilson    | East    | Home & Kitchen    | Non-Stick Pan     | 1 | 8888 Completed    | Credit Card | 1 | 8888    |
| 82 | ORD102699 | 2024-01-20 | CUST1454 | Fajar Wijaya   | Midwest | Clothing          | Hoodie            | 1 | 8888 Completed    | Gift Card   | 1 | 8888    |
| 83 | ORD102755 | 2023-11-10 | CUST1809 | Kevin Smith    | Midwest | Books             | Business Strategy | 2 | 4999.99 Completed | PayPal      | 2 | 4999.99 |
| 84 | ORD101880 | 2024-05-06 | CUST1535 | Maria Brown    | Midwest | Toys              | Action Figure     | 1 | 4999.99 Cancelled | Credit Card | 1 | 4999.99 |
| 85 | ORD100921 | 2024-05-15 | CUST1725 | Nina Pratama   | West    | Books             | Business Strategy | 2 | 1234.56 Completed | Credit Card | 2 | 1234.56 |
| 86 | ORD101822 | 2024-11-14 | CUST1564 | Maria Chen     | South   | Sports & Outdoors | Jump Rope         | 2 | 1234.56 Completed | Credit Card | 2 | 1234.56 |
| 87 | ORD101110 | 2024-07-10 | CUST1842 | Siti Salim     | East    | Books             | Cookbook          | 2 | 1234.56 Completed |             | 2 | 1234.56 |
| 88 | ORD101599 | 2024-12-06 | CUST1138 | Fajar Anderson | Midwest | Electronics       | Power Bank        | 1 | 1234.56 Completed | Gift Card   | 1 | 1234.56 |
| 89 | ORD101460 | 2024-01-01 | CUST1145 | Linda Smith    | Midwest | Clothing          | Wool Socks        | 1 | 1234.56 completed | Debit Card  | 1 | 1234.56 |
| 90 | ORD100829 | 2023-02-16 | CUST1399 | Rina Lee       | Midwest | Sports & Outdoors | Yoga Mat          | 1 | 1234.56 Completed | PayPal      | 0 | 1234.56 |
| 91 | ORD101912 | 2024-07-21 | CUST1634 | James Sari     | East    | Electronics       | Power Bank        | 2 | 149.71 Pending    | Debit Card  | 2 | 149.71  |

Gambar 17. 12 baris dengan unit\_price ekstrem (8888 / 4999,99 / 1234,56) diseleksi untuk dihapus

- Unit price: 12 baris dengan nilai 8888 / 4999,99 / 1234,56 dihapus. Nilai-nilai ini tidak sesuai kategori produknya (contoh: Puzzle 1000pc seharga \$8.888), dan nilai 1234,56 muncul identik berulang di 6 produk berbeda — indikasi kuat error sistem, bukan variasi harga wajar.

|      | A         | B          | C           | D               | E        | F                 | G                 | H         | I           | J            | K              | L              | M          | N |
|------|-----------|------------|-------------|-----------------|----------|-------------------|-------------------|-----------|-------------|--------------|----------------|----------------|------------|---|
| 1    | order_id  | order_date | customer_id | customer_name   | region   | category          | product_name      | quantity  | unit_price  | order_status | payment_method | quantity_clean | unit_price |   |
| 3939 | ORD101018 | 2023-03-11 | CUST1260    | Fajar Pratama   | East     | Sports & Outdoors | Resistance Bands  | 1         | Completed   | Gift Card    |                | 1              |            |   |
| 3939 | ORD100917 | 2024-03-21 | CUST1394    | Maria Hidayat   | South    | Sports & Outdoors | Hiking Backpack   | 1         | Returned    | PayPal       |                | 1              |            |   |
| 3939 | ORD102799 | 2024-05-22 | CUST1762    | Rina Nugroho    | South    | Sports & Outdoors | Resistance Bands  | 1         | Completed   | Credit Card  |                | 1              |            |   |
| 3939 | ORD100944 | 2024-10-16 | CUST1380    | Lisa Brown      | West     | Beauty            | Sunscreen SPF50   | 1         | Completed   | PayPal       |                | 1              |            |   |
| 3939 | ORD101530 | 2024-04-30 | CUST1287    | John Pratama    | South    | Sports & Outdoors | Yoga Mat          | 1         | Pending     | PayPal       |                | 1              |            |   |
| 3939 | ORD101879 | 2024-04-05 | CUST1136    | Budi Hidayat    | Midwest  | Books             | Self-Help Guide   | 1         | Completed   | Debit Card   |                | 1              |            |   |
| 3940 | ORD103237 | 2023-10-11 | CUST1624    | Budi Smith      | South    | Home & Kitchen    | Cutting Board     | 1         | Completed   | Gift Card    |                | 1              |            |   |
| 3940 | ORD100905 | 2024-05-15 | CUST1350    | Linda Salim     | Midwest  | Clothing          | Hoodie            | 1         | Completed   | Credit Card  |                | 1              |            |   |
| 3940 | ORD103913 | 2023-01-24 | CUST1026    | Putri Yusuf     | Midwest  | Home & Kitchen    | Cutting Board     | 1         | Completed   | Credit Card  |                | 1              |            |   |
| 3940 | ORD101125 | 2023-01-27 | CUST1516    | Fajar Brown     | Midwest  | Electronics       | Webcam            | 1         | Returned    | Gift Card    |                | 1              |            |   |
| 3940 | ORD103970 | 2023-04-26 | CUST1329    | Andi Tan        | South    | Clothing          | Denim Jeans       | 1         | Returned    | PayPal       |                | 1              |            |   |
| 3940 | ORD101487 | 2023-08-21 | CUST1687    | Siti Sari       | South    | Toys              | Action Figure     | 1         | Completed   | Credit Card  |                | 1              |            |   |
| 3940 | ORD101406 | 2024-08-14 | CUST1349    | John Halim      | East     | Toys              | Board Game        | 1         | Completed   | Gift Card    |                | 1              |            |   |
| 3940 | ORD102133 | 2024-11-30 | CUST1745    | Ahmad Halim     | East     | Clothing          | Denim Jeans       | 1         | Completed   | Debit Card   |                | 1              |            |   |
| 3940 | ORD103818 | 2023-09-09 | CUST1660    | Andi Salim      | East     | Sports & Outdoors | Hiking Backpack   | 1         | Completed   | Credit Card  |                | 1              |            |   |
| 3940 | ORD100019 | 2024-08-06 | CUST1412    | Siti Anderson   | South    | Home & Kitchen    | Coffee Maker      | 1         | Completed   | Debit Card   |                | 1              |            |   |
| 3950 | ORD101245 | 2024-04-24 | CUST1412    | Siti Anderson   | West     | Clothing          | Wool Socks        | 1         | Cancelled   | PayPal       |                | 1              |            |   |
| 3950 | ORD103531 | 2024-02-27 | CUST1894    | Putri Anderson  | East     | Home & Kitchen    | Blender           | 1         | Completed   | Gift Card    |                | 1              |            |   |
| 3950 | ORD101540 | 2024-06-01 | CUST1754    | Kevin Salim     | West     | Electronics       | Power Bank        | 1         | Returned    | Debit Card   |                | 1              |            |   |
| 3950 | ORD100027 | 2023-05-05 | CUST1324    | Budi Santos     | Midwest  | Sports & Outdoors | Resistance Bands  | 1         | Completed   | Gift Card    |                | 1              |            |   |
| 3950 | ORD102052 | 2024-11-04 | CUST1837    | Devil Johnson   | Midwest  | Home & Kitchen    | Blender           | 1         | Completed   | Debit Card   |                | 1              |            |   |
| 3950 | ORD103958 | 2024-03-30 | CUST1005    | Devil Wijaya    | Toys     | Building Blocks   | 1                 | Completed | Gift Card   |              | 1              |                |            |   |
| 3950 | ORD100155 | 2023-08-18 | CUST1168    | Linda Nugroho   | East     | Toys              | Board Game        | 1         | Completed   | PayPal       |                | 1              |            |   |
| 3950 | ORD103247 | 2024-11-26 | CUST1177    | Ayu Pratama     | East     | Beauty            | Sunscreen SPF50   | 1         | Completed   | Debit Card   |                | 1              |            |   |
| 3950 | ORD102778 | 2023-04-08 | CUST1161    | Ayu Yusuf       | East     | Books             | Business Strategy | 1         | Completed   | Credit Card  |                | 1              |            |   |
| 3950 | ORD103476 | 2024-10-03 | CUST1171    | James Anderson  | Midwest  | Books             | Children's Book   | 1         | Completed   | PayPal       |                | 1              |            |   |
| 3950 | ORD100166 | 2024-06-30 | CUST1846    | Kevin Kurniawan | East     | Books             | Business Strategy | 1         | Returned    | Gift Card    |                | 1              |            |   |
| 3950 | ORD100607 | 2023-07-01 | CUST1617    | Robert Tan      | South    | Beauty            | Shampoo           | 1         | Pending     | Debit Card   |                | 1              |            |   |
| 3960 | ORD101989 | 2023-08-31 | CUST1684    | Fajar Salim     | East     | Toys              | Puzzle 1000pc     | 1         | Completed   | PayPal       |                | 1              |            |   |
| 3960 | ORD103234 | 2024-09-18 | CUST1642    | South           | Clothing | Hoodie            | 1                 | Returned  | Gift Card   |              | 1              |                |            |   |
| 3960 | ORD101718 | 2024-11-30 | CUST1256    | James Wijaya    | West     | Clothing          | Winter Jacket     | 1         | Completed   | Credit Card  |                | 1              |            |   |
| 3960 | ORD101526 | 2023-10-22 | CUST1811    | Putri Santos    | Midwest  | Clothing          | Denim Jeans       | 1         | Completed   | PayPal       |                | 1              |            |   |
| 3960 | ORD100898 | 2023-05-02 | CUST1575    | West            | Clothing | Denim Jeans one   | Completed         | Gift Card |             | 1            |                |                |            |   |
| 3960 | ORD103206 | 2024-10-17 | CUST1417    | Electronics     | Webcam   | 1                 | Completed         | PayPal    |             | 1            |                |                |            |   |
| 3960 | ORD103339 | 2024-11-25 | CUST1154    | Michael Tan     | East     | Electronics       | Laptop Stand      | 1         | Completed   | PayPal       |                | 1              |            |   |
| 3960 | ORD100759 | 2023-11-14 | CUST1252    | Ayu Brown       | West     | Home & Kitchen    | Throw Pillow      | 1         | Completed   | Debit Card   |                | 1              |            |   |
| 3970 | ORD103614 | 2024-08-27 | CUST1386    | Lisa Johnson    | Midwest  | Books             | Business Strategy | 1         | Completed   | Credit Card  |                | 1              |            |   |
| 3970 | ORD102400 | 2024-07-10 | CUST1324    | Budi Santos     | Midwest  | Toys              | Action Figure     | 1         | Completed   | Debit Card   |                | 1              |            |   |
| 3970 | ORD101194 | 2024-07-13 | CUST1705    | Ayu Brown       | Midwest  | Electronics       | Smartphone Case   | Completed | Credit Card |              | 0              |                |            |   |
| 3970 | ORD100216 | 2024-09-12 | CUST1356    | Ahmad Tan       | Midwest  | Home & Kitchen    | Throw Pillow      | Pending   | Gift Card   |              | 0              |                |            |   |
| 3970 | ORD101914 | 2023-09-23 | CUST1838    | Budi Chen       | South    | Home & Kitchen    | Throw Pillow      | Completed | PayPal      |              | 0              |                |            |   |
| 3970 | ORD100143 | 2024-10-24 | CUST1799    | Robert Hidayat  | South    | Toys              | Building Blocks   | Completed | Credit Card |              | 0              |                |            |   |

Gambar 18. Sebelum: outlier quantity/unit\_price masih tercampur di dalam data

| #  | A         | B          | C           | D                 | E                 | F                 | G                 | H        | I          | J            | K              | L              | M                |
|----|-----------|------------|-------------|-------------------|-------------------|-------------------|-------------------|----------|------------|--------------|----------------|----------------|------------------|
| 1  | order_id  | order_date | customer_id | customer_name     | region            | category          | product_name      | quantity | unit_price | order_status | payment_method | quantity_clean | unit_price_clean |
| 7  | ORD100006 | 2024-01-16 | CUST1580    | Linda Tan         | South             | Electronics       | Smartphone Case   | 1        | 81.85      | Completed    | Credit Card    | 1              | 81.85            |
| 8  | ORD100007 | 2023-09-26 | CUST1753    | Kevin Lee         | South             | Clothing          | Running Shorts    | 1        | 77.87      | Completed    | Gift Card      | 1              | 77.87            |
| 9  | ORD100008 | 2024-11-29 | CUST1075    | Linda Anderson    | South             | Sports & Outdoors | Water Bottle      | 1        | 43.76      | Pending      | Debit Card     | 1              | 43.76            |
| 10 | ORD100009 | 2023-05-05 | CUST1726    | John Tan          | Midwest           | Beauty            | Shampoo           | 1        | 36.12      | Pending      | PayPal         | 1              | 36.12            |
| 11 | ORD100010 | 2024-02-12 | CUST1132    | Robert Tan        | East              | Books             | Fiction Novel     | 1        | 10.00      | Pending      | Gift Card      | 1              | 10               |
| 12 | ORD100011 | 2024-07-18 | CUST1460    | Rina Sari         | Midwest           | Sports & Outdoors | Jump Rope         | 3        | 36.61      | Completed    | PayPal         | 3              | 36.61            |
| 13 | ORD100012 | 2023-12-21 | CUST1311    | John Nugroho      | Clothing          | Denim jeans       | 3                 | 18.37    | Returned   | Debit Card   | 3              | 18.37          |                  |
| 14 | ORD100013 | 2024-11-05 | CUST1546    | John Garcia       | West              | Electronics       | Smartphone Case   | 1        | 72.72      | Completed    | Debit Card     | 1              | 72.72            |
| 15 | ORD100014 | 1900-01-00 | CUST1590    | Putri Tan         | West              | Home & Kitchen    | LED Desk Lamp     | 3        | 93.60      | Completed    | Gift Card      | 3              | 93.6             |
| 16 | ORD100015 | 2023-06-11 | CUST1515    | Sarah Salim       | East              | Home & Kitchen    | Cutting Board     | 4        | 106.12     | Completed    | Gift Card      | 4              | 106.12           |
| 17 | ORD100016 | 2023-11-23 | CUST1353    | Andi Halim        | Sports & Outdoors | Water Bottle      | 2                 | 28.97    | Completed  | Credit Card  | 2              | 28.97          |                  |
| 18 | ORD100017 | 2023-07-14 | CUST1254    | Nina Garcia       | Midwest           | Toys              | Puzzle 1000pc     | 1        | 25.48      | Completed    | Gift Card      | 1              | 25.48            |
| 19 | ORD100018 | 2023-02-21 | CUST1468    | Michael Tan       | Midwest           | Clothing          | Denim jeans       | 1        | 35.43      | Completed    | Gift Card      | 1              | 35.43            |
| 20 | ORD100019 | 2024-08-06 | CUST1412    | Siti Anderson     | South             | Home & Kitchen    | Coffee Maker      | 1        |            | Completed    | Debit Card     | 1              |                  |
| 21 | ORD100020 | 2024-02-08 | CUST1696    | David Hidayat     | West              | Books             | Fiction Novel     | 1        | 28.56      | Completed    | Debit Card     | 1              | 28.56            |
| 22 | ORD100021 | 2024-06-15 | CUST1039    | Robert Tan        | Midwest           | Clothing          | Wool Socks        | 1        | 54.35      | Completed    | PayPal         | 1              | 54.35            |
| 23 | ORD100022 | 2024-01-05 | CUST1344    | Ahmad Salim       | South             | Books             | Business Strategy | 1        | 14.12      | Completed    | Debit Card     | 1              | 14.12            |
| 24 | ORD100023 | 2023-02-22 | CUST1644    | Andi Salim        | Midwest           | Electronics       | Wireless Mouse    | 2        | 67.21      | Completed    | PayPal         | 2              | 67.21            |
| 25 | ORD100024 | 2024-03-27 | CUST1586    | Putri Smith       | West              | Beauty            | Moisturizer       | 1        | 9.63       | Completed    | PayPal         | 1              | 9.63             |
| 26 | ORD100025 | 2024-11-09 | CUST1744    | Andi Sari         | West              | Electronics       | Smartphone Case   | 3        | 100.47     | Completed    | Gift Card      | 3              | 100.47           |
| 27 | ORD100026 | 2023-09-10 | CUST1738    | Fajar Halim       | South             | Beauty            | Face Mask Set     | 2        | 9.96       | Completed    | Debit Card     | 2              | 9.96             |
| 28 | ORD100027 | 2023-05-05 | CUST1324    | Budi Santos       | Midwest           | Sports & Outdoors | Resistance Bands  | 1        |            | Completed    | Gift Card      | 1              |                  |
| 29 | ORD100028 | 2024-05-05 | CUST1731    | Robert Anderson   | South             | Electronics       | Laptop Stand      | 1        | 115.78     | Completed    | Debit Card     | 1              | 115.78           |
| 30 | ORD100029 | 2023-04-13 | CUST1685    | Budi Kurniawan    | West              | Books             | Cookbook          | 1        | 25.04      | Completed    | Gift Card      | 1              | 25.04            |
| 31 | ORD100030 | 2023-10-22 | CUST1416    |                   | South             | Beauty            | Lip Balm          | 1        | 31.17      | Completed    | Credit Card    | 1              | 31.17            |
| 32 | ORD100032 | 2024-05-31 | CUST1656    | Ayu Yusuf         | South             | Books             | Business Strategy | 4        | 16.56      | Completed    |                | 4              | 16.56            |
| 33 | ORD100033 | 2024-09-17 | CUST1345    | Michael Santos    | West              | Toys              | RC Car            | 4        | 35.43      | Cancelled    | Gift Card      | 4              | 35.43            |
| 34 | ORD100034 | 2023-08-30 | CUST1331    | Ayu Chen          | Midwest           | Clothing          | Cotton T-Shirt    | 3        | 58.79      | Completed    | Debit Card     | 3              | 58.79            |
| 35 | ORD100035 | 2023-06-13 | CUST1063    | David Anderson    | South             | Beauty            | Moisturizer       | 3        | 33.38      | Completed    | Debit Card     | 3              | 33.38            |
| 36 | ORD100036 | 2024-12-19 | CUST1489    | Nina Salim        | Midwest           | Clothing          | Running Shorts    | 3        | 36.29      | Completed    | Gift Card      | 3              | 36.29            |
| 37 | ORD100037 | 1900-01-00 | CUST1806    | Ahmad Hidayat     | Midwest           | Electronics       | Wireless Mouse    | 1        | 94.76      | Completed    | PayPal         | 1              | 94.76            |
| 38 | ORD100038 | 2024-04-09 | CUST1597    | Ahmad Nugroho     | South             | Sports & Outdoors | Yoga Mat          | 2        | 12.75      | Completed    | Gift Card      | 2              | 12.75            |
| 39 | ORD100039 | 2024-04-09 | CUST1836    | Kevin Salim       | East              | Electronics       | USB-C Cable       | 2        | 146.74     | Completed    | Credit Card    | 2              | 146.74           |
| 40 | ORD100040 | 2024-01-27 | CUST1184    | Michael Kurniawan | South             | Toys              | Puzzle 1000pc     | 1        | 21.76      | Completed    | Gift Card      | 1              | 21.76            |
| 41 | ORD100041 | 2023-01-04 | CUST1182    | Dewi Tan          | West              | Books             | Children's Book   | 0        | 27.92      | Completed    | Debit Card     | 0              | 27.92            |
| 42 | ORD100042 | 2024-07-11 | CUST1158    | Nina Nugroho      | East              | Home & Kitchen    | Non-Stick Pan     | 1        | 93.85      | Completed    | Credit Card    | 1              | 93.85            |
| 43 | ORD100043 | 2023-08-28 | CUST1250    | Linda Smith       | South             | Beauty            | Moisturizer       | 3        | 12.75      | Completed    | Credit Card    | 3              | 12.75            |
| 44 | ORD100044 | 2023-08-27 | CUST1391    | Fajar Wilson      | West              | Books             | Self-Help Guide   | 1        | 24.90      | Pending      | PayPal         | 1              | 24.9             |
| 45 | ORD100045 | 2023-12-19 | CUST1695    | James Johnson     | South             | Sports & Outdoors | Hiking Backpack   | 2        | 10.44      | Returned     | Gift Card      | 2              | 10.44            |
| 46 | ORD100046 | 2024-12-25 | CUST1484    | James Anderson    | West              | Clothing          | Baseball Cap      | 1        | 42.08      | Completed    | Gift Card      | 1              | 42.08            |

Gambar 19. Sesudah: kolom quantity dan unit\_price bersih, tanpa nilai ekstrem

| Hari  | Fase                | Task                                            | Tools | Selesai? |
|-------|---------------------|-------------------------------------------------|-------|----------|
| Day 1 | Excel Data Cleaning | Ambil dataset e-commerce sales                  | Excel | SELESAI  |
| Day 1 | Excel Data Cleaning | Cek & tangani missing values                    | Excel |          |
| Day 1 | Excel Data Cleaning | Cek & hapus duplicate records                   | Excel | SELESAI  |
| Day 1 | Excel Data Cleaning | Perbaiki data types yang salah                  | Excel | SELESAI  |
| Day 1 | Excel Data Cleaning | Standarkan kategori yang tidak konsisten        | Excel | SELESAI  |
| Day 1 | Excel Data Cleaning | Format ulang kolom tanggal                      | Excel | SELESAI  |
| Day 1 | Excel Data Cleaning | Cek outlier                                     | Excel | SELESAI  |
| Day 1 | Excel Data Cleaning | Buat calculated columns                         | Excel |          |
| Day 1 | Excel Data Cleaning | Dokumentasikan raw -> cleaning -> clean dataset | Excel |          |
| Day 1 | Excel Data Cleaning | Simpan sebagai cleaned_sales_data.xlsx          | Excel |          |

Gambar 20. Checklist Day 1 setelah tahap perbaikan tipe data dan outlier selesai

### 3.5 Penanganan Missing Value

Setelah tahap sebelumnya, enam kolom masih memiliki missing value dengan jumlah dan tingkat kekritisannya berbeda-beda. Jumlah missing value per kolom dihitung menggunakan COUNTBLANK di area kosong di luar tabel:

```
=COUNTBLANK(D2:D3975) {ganti huruf kolom sesuai field yang dicek}
```

| SUM  | :         | X          | ✓           | f_x            | =COUNTBLANK(D2:D3975) |
|------|-----------|------------|-------------|----------------|-----------------------|
| 1    | order_id  | order_date | customer_id | customer_name  | region                |
| 3963 | ORD103988 | 2024-04-15 | CUST1412    | Siti Anderson  | West                  |
| 3964 | ORD103989 | 2023-06-18 | CUST1696    | David Hidayat  | West                  |
| 3965 | ORD103990 | 2024-12-01 | CUST1774    | Maria Wijaya   | West                  |
| 3966 | ORD103991 | 2023-11-10 | CUST1664    | Dewi Nugroho   | Midwest               |
| 3967 | ORD103992 | 2023-04-25 | CUST1292    | Budi Halim     | South                 |
| 3968 | ORD103993 | 2024-03-03 | CUST1382    | David Hidayat  | Midwest               |
| 3969 | ORD103994 | 2024-01-17 | CUST1058    | John Johnson   | West                  |
| 3970 | ORD103995 | 2023-02-16 | CUST1113    | Dewi Nugroho   | Midwest               |
| 3971 | ORD103996 | 2023-02-05 | CUST1248    | Andi Pratama   | East                  |
| 3972 | ORD103997 | 2024-02-22 | CUST1494    | Robert Pratama | East                  |
| 3973 | ORD103998 | 2023-11-05 | CUST1348    | Putri Nugroho  | East                  |
| 3974 | ORD103999 | 2023-01-20 | CUST1421    | Ayu Lee        | South                 |
| 3975 | ORD104000 | 2024-02-12 | CUST1580    | Linda Tan      | Midwest               |
| 3976 |           |            |             |                |                       |
| 3977 |           |            |             |                |                       |
| 3978 |           |            |             |                |                       |
| 3979 |           |            |             |                |                       |
| 3980 |           |            |             | customer_name  |                       |
| 3981 |           |            |             | D3975)         |                       |
| 3982 |           |            |             |                |                       |
| 3983 |           |            |             |                |                       |
| 3984 |           |            |             |                |                       |
| 3985 |           |            |             |                |                       |

Gambar 21. Formula COUNTBLANK diterapkan di luar tabel untuk menghitung missing value tiap kolom

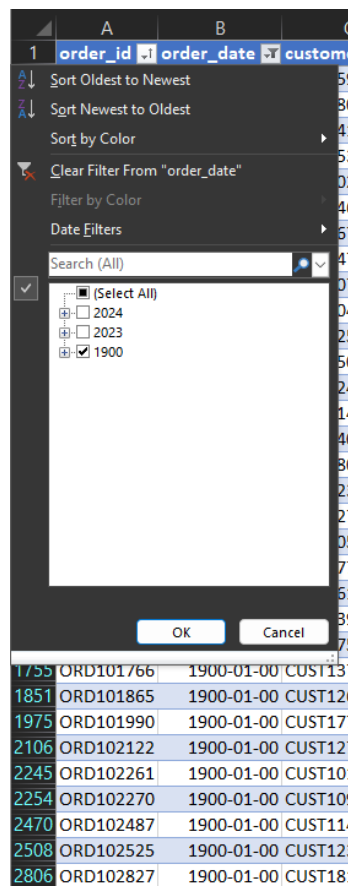
|                |  |     |
|----------------|--|-----|
| customer_name  |  | 120 |
| region         |  | 79  |
| payment_method |  | 158 |
| order_date     |  | 0   |
| quantity       |  | 0   |
| unit_price     |  | 77  |

Gambar 22. Hasil rekap awal — order\_date sempat terbaca 0 karena nilai kosong tersembunyi sebagai angka 0

|                |  |     |
|----------------|--|-----|
| customer_name  |  | 120 |
| region         |  | 79  |
| payment_method |  | 158 |
| order_date     |  | 0   |
| quantity       |  | 59  |
| unit_price     |  | 77  |

Gambar 23. Rekap setelah quantity yang sempat tersamar sebagai 0 dikembalikan menjadi blank (naik menjadi 59)

order\_date sempat terbaca 0 missing, padahal ada nilai yang tersembunyi sebagai angka 0 (bekas sel kosong yang diformat jadi tanggal 1 Januari 1900). Ditemukan lewat filter tahun pada kolom order\_date:



Gambar 24. Filter order\_date menemukan baris dengan tahun 1900 — bekas sel kosong yang tersembunyi

|                |     |
|----------------|-----|
| customer_name  | 120 |
| region         | 79  |
| payment_method | 158 |
| order_date     | 40  |
| quantity       | 59  |
| unit_price     | 77  |

Gambar 25. Rekap final missing value per kolom setelah semua nilai tersembunyi ditemukan

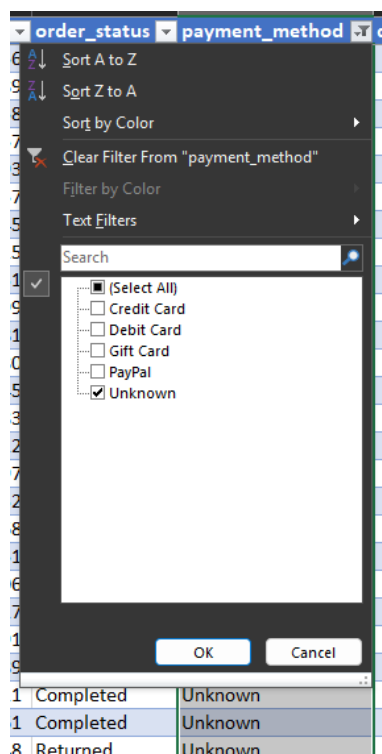
Strategi penanganan disesuaikan per kolom — kolom kategorikal diisi label "Unknown", kolom yang dipakai untuk perhitungan (quantity, unit\_price) diisi menggunakan estimasi statistik, sedangkan order\_date dihapus karena tidak dapat direkonstruksi secara valid.

| Kolom          | Jumlah | %    | Strategi Penanganan                                                                                              |
|----------------|--------|------|------------------------------------------------------------------------------------------------------------------|
| payment_method | 158    | 4,0% | Diisi "Unknown" — kolom kategori, tidak memengaruhi perhitungan.                                                 |
| customer_name  | 120    | 3,0% | Diisi "Unknown Customer" — identitas tetap tersedia via customer_id.                                             |
| region         | 79     | 2,0% | Direcovery dari transaksi lain milik customer_id yang sama (INDEX/MATCH array formula); sisanya diisi "Unknown". |
| unit_price     | 77     | 1,9% | Diisi rata-rata harga per product_name (AVERAGEIFS) — lebih akurat dibanding rata-rata global.                   |
| quantity       | 59     | 1,5% | Diisi median quantity keseluruhan (1 unit) — median dipilih karena data condong ke angka kecil.                  |
| order_date     | 40     | 1,0% | Baris dihapus — tanggal transaksi krusial untuk analisis tren dan tidak dapat direkonstruksi secara valid.       |

Tabel 2. Ringkasan missing value dan strategi penanganannya

### payment\_method dan customer\_name

Kedua kolom kategorikal ini diisi memakai filter (Blanks) lalu Select All + ketik nilai pengganti + Alt+Enter untuk mengisi seluruh sel yang terseleksi sekaligus:



Gambar 26. Filter payment\_method menampilkan baris kosong, siap diisi "Unknown"

**region — dicoba direcovery lebih dulu**

Kolom region dicoba direcovery dari transaksi lain milik customer\_id yang sama sebelum jatuh ke "Unknown", menggunakan formula array (ditekan dengan Ctrl+Shift+Enter):

```
=IFERROR(INDEX($region$2:$region$3975, MATCH(1,
($customer_id$2:$customer_id$3975=[@customer_id]) * ($region$2:$region$3975<>""), 0)),
"") {Ctrl+Shift+Enter}
```

|    | A         | B          | C           | D                | E      | F                 | G                 | H        | I          | J            | K              | L                |
|----|-----------|------------|-------------|------------------|--------|-------------------|-------------------|----------|------------|--------------|----------------|------------------|
|    | order_id  | order_date | customer_id | customer_name    | region | category          | product_name      | quantity | unit_price | order_status | payment_method | region_recovered |
| 2  | ORD100553 | 2024-11-04 | CUST1305    | Linda Nugroho    |        | Home & Kitchen    | Storage Container | 1        | 31.12      | Completed    | Debit Card     | East             |
| 3  | ORD101005 | 2023-04-05 | CUST1755    | Budi Sari        |        | Electronics       | Smartphone Case   | 1        | 29.75      | Returned     | PayPal         | East             |
| 4  | ORD101116 | 2023-11-02 | CUST1733    | Lisa Sari        |        | Beauty            | Face Mask Set     | 1        | 29.15      | Cancelled    | Debit Card     | East             |
| 5  | ORD101125 | 2023-01-27 | CUST1516    | Fajar Brown      |        | Electronics       | Webcam            | 1        |            | Returned     | Gift Card      | East             |
| 6  | ORD101190 | 2023-03-17 | CUST1638    | Maria Kurniawan  |        | Toys              | Building Blocks   | 1        | 29.17      | Completed    | Credit Card    | East             |
| 7  | ORD101319 | 2023-01-24 | CUST1707    | Michael Yusuf    |        | Electronics       | Power Bank        | 3        | 70.28      | Completed    | Debit Card     | East             |
| 8  | ORD101435 | 2023-05-15 | CUST1844    | Ahmad Johnson    |        | Clothing          | Baseball Cap      | 5        | 59.21      | Completed    | PayPal         | East             |
| 9  | ORD101524 | 2024-01-18 | CUST1172    | Nina Lee         |        | Home & Kitchen    | Blender           | 1        | 78.32      | Completed    | Gift Card      | East             |
| 10 | ORD101662 | 2024-04-08 | CUST1121    | David Nugroho    |        | Home & Kitchen    | Storage Container | 1        | 24.34      | Completed    | PayPal         | East             |
| 11 | ORD102055 | 2024-08-04 | CUST1708    | Ayu Smith        |        | Beauty            | Face Mask Set     | 2        | 32.63      | Completed    | Gift Card      | East             |
| 12 | ORD102156 | 2024-01-11 | CUST1798    | Siti Anderson    |        | Beauty            | Lip Balm          | 1        | 6.38       | Completed    | Gift Card      | East             |
| 13 | ORD102170 | 2023-10-16 | CUST1561    | Nina Santos      |        | Toys              | Building Blocks   | 2        | 41.55      | Completed    | Debit Card     | East             |
| 14 | ORD102589 | 2024-05-25 | CUST1558    | Sarah Wilson     |        | Sports & Outdoors | Hiking Backpack   | 2        | 56.86      | Completed    | Credit Card    | East             |
| 15 | ORD102759 | 2024-12-22 | CUST1572    | Maria Yusuf      |        | Toys              | Board Game        | 2        | 52.88      | Cancelled    | Credit Card    | East             |
| 16 | ORD102912 | 2024-03-14 | CUST1851    | Robert Salim     |        | Toys              | Action Figure     | 4        | 33.78      | Completed    | Gift Card      | East             |
| 17 | ORD103206 | 2024-10-17 | CUST1417    | Unknown Customer |        | Electronics       | Webcam            | 1        |            | Completed    | PayPal         | East             |
| 18 | ORD103366 | 2024-11-14 | CUST1723    | Ayu Wijaya       |        | Home & Kitchen    | LED Desk Lamp     | 2        | 43.94      | Completed    | Debit Card     | East             |
| 19 | ORD103436 | 2023-07-04 | CUST1402    | Rina Nugroho     |        | Clothing          | Wool Socks        | 1        | 33.39      | Completed    | PayPal         | East             |

Gambar 27. Kolom region\_recovered berhasil menemukan region dari transaksi lain customer yang sama

**unit\_price dan quantity — diisi estimasi statistik**

Kolom unit\_price diisi memakai AVERAGEIFS — rata-rata harga dihitung khusus dari produk yang sama (product\_name), bukan rata-rata seluruh dataset:

```
=IF([@unit_price]="", AVERAGEIFS($unit_price$2:$unit_price$3975,
$product_name$2:$product_name$3975, [@product_name]), [@unit_price])
```

| unit_price | order_status | payment_method | unit_price_fill |
|------------|--------------|----------------|-----------------|
|            | Completed    | Credit Card    | 83.09465909     |
|            | Pending      | Gift Card      | 63.9183908      |
|            | Completed    | Credit Card    | 36.80701923     |
|            | Completed    | PayPal         | 63.9183908      |

Gambar 28. Kolom unit\_price\_fill — harga kosong terisi rata-rata harga produk yang sama

Kolom quantity diisi memakai MEDIAN (bukan AVERAGE), karena data quantity condong ke angka kecil (1–5) sehingga median lebih tahan terhadap distorsi dibanding rata-rata:

```
=IF([@quantity]="", MEDIAN($quantity$2:$quantity$3975), [@quantity])
```

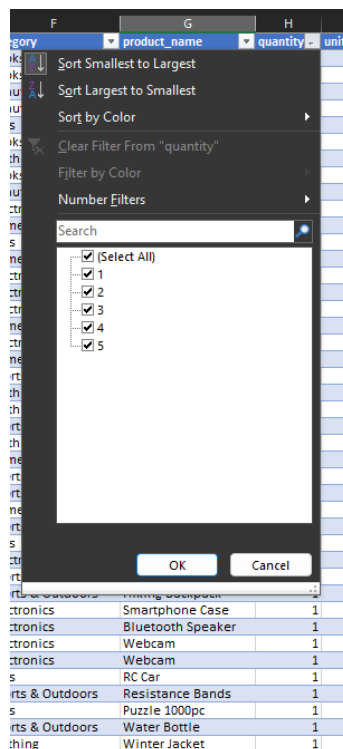
| quantity | unit_price | order_status | payment_method | unit_price_fill | quantity_fill |
|----------|------------|--------------|----------------|-----------------|---------------|
| 21.69    | Completed  | Credit Card  |                | 21.69           | 1             |
| 11.03    | Completed  | Gift Card    |                | 11.03           | 1             |
| 58.91    | Completed  | Credit Card  |                | 58.91           | 1             |
| 122.70   | Completed  | Debit Card   |                | 122.7           | 1             |
| 17.39    | Completed  | Credit Card  |                | 17.39           | 1             |
| 49.25    | Returned   | Credit Card  |                | 49.25           | 1             |
| 13.42    | Completed  | Debit Card   |                | 13.42           | 1             |
| 64.16    | Completed  | Unknown      |                | 64.16           | 1             |
| 33.37    | Completed  | Gift Card    |                | 33.37           | 1             |
| 58.85    | Completed  | Gift Card    |                | 58.85           | 1             |
| 19.42    | Completed  | Debit Card   |                | 19.42           | 1             |
| 76.05    | Completed  | Debit Card   |                | 76.05           | 1             |
| 75.06    | Completed  | Debit Card   |                | 75.06           | 1             |
|          | Completed  | Credit Card  |                | 83.09465909     | 1             |
| 17.66    | Completed  | Debit Card   |                | 17.66           | 1             |

Gambar 29. Kolom unit\_price\_fill dan quantity\_fill tampil bersama — quantity kosong konsisten terisi median (1)



Catatan cepat — sintaks dasar ketiga fungsi ini:

- **AVERAGE(range)** — rata-rata semua angka di range, tanpa syarat.
- **AVERAGEIFS(average\_range, criteria\_range1, criteria1, ...)** — rata-rata dari average\_range, tapi hanya baris yang memenuhi syarat (bisa lebih dari satu syarat).
- **MEDIAN(range)** — nilai tengah dari sekumpulan angka; lebih tahan terhadap outlier dibanding AVERAGE.



Gambar 30. Filter quantity setelah semua missing value tertangani — hanya tersisa nilai wajar 1–5

## Menggabungkan hasil kembali ke kolom asli

Saat memindahkan hasil kolom bantu (seperti `region_recovered`) kembali ke kolom asli, cara yang lebih efisien untuk baris dalam jumlah banyak adalah mengurutkan (Sort A-Z) kolom bantu terlebih dahulu — baris yang berhasil direcovery akan berkumpul jadi satu grup, sehingga bisa disalin sekaligus per grup, bukan satu per satu:

|      | A         | B          | C           | D                 | E       | F                 | G                | H        | I          | J            | K              | L               | M             |
|------|-----------|------------|-------------|-------------------|---------|-------------------|------------------|----------|------------|--------------|----------------|-----------------|---------------|
|      | order_id  | order_date | customer_id | customer_name     | region  | category          | product_name     | quantity | unit_price | order_status | payment_method | unit_price_fill | quantity_fill |
| 29   | ORD103119 |            | CUST1171    | James Anderson    | East    | Beauty            | Shampoo          | 1        | 8.19       | Completed    | PayPal         | 8.19            | 1             |
| 333  | ORD101366 |            | CUST1407    | Rina Smith        | East    | Books             | Children's Book  | 2        | 13.80      | Returned     | PayPal         | 13.80           | 2             |
| 586  | ORD100368 |            | CUST1472    | Andi Halim        | Midwest | Beauty            | Shampoo          | 2        | 17.57      | Completed    | Gift Card      | 17.57           | 2             |
| 65   | ORD103834 |            | CUST1853    | Robert Smith      | East    | Beauty            | Face Mask Set    | 2        | 18.12      | Completed    | Credit Card    | 18.12           | 2             |
| 830  | ORD102827 |            | CUST1818    | Siti Salim        | East    | Toys              | RC Car           | 1        | 21.00      | Completed    | Credit Card    | 21.00           | 1             |
| 845  | ORD102261 |            | CUST1016    | John Pratama      | Midwest | Beauty            | Lip Balm         | 1        | 21.12      | Completed    | Debit Card     | 21.12           | 1             |
| 947  | ORD101212 |            | CUST1503    | Andi Anderson     | East    | Beauty            | Sunscreen SPF50  | 1        | 22.65      | Returned     | Gift Card      | 22.65           | 1             |
| 987  | ORD100583 |            | CUST1076    | Kevin Santos      | East    | Clothing          | Running Shorts   | 2        | 23.24      | Completed    | PayPal         | 23.24           | 2             |
| 1058 | ORD100121 |            | CUST1538    | Dewi Kurniawan    | East    | Toys              | RC Car           | 1        | 24.15      | Returned     | Gift Card      | 24.15           | 1             |
| 1074 | ORD100824 |            | CUST1040    | Lisa Garcia       | East    | Toys              | Puzzle 1000pc    | 2        | 24.38      | Cancelled    | PayPal         | 24.38           | 2             |
| 1088 | ORD101865 |            | CUST1287    | Maria Anderson    | East    | Home & Kitchen    | Blender          | 2        | 24.78      | Completed    | PayPal         | 24.78           | 2             |
| 1378 | ORD100265 |            | CUST1023    | Siti Wilson       | East    | Home & Kitchen    | Non-Stick Pan    | 1        | 28.43      | Completed    | Debit Card     | 28.43           | 1             |
| 1500 | ORD101196 |            | CUST1254    | Nina Garcia       | East    | Beauty            | Sunscreen SPF50  | 2        | 30.21      | returned     | Gift Card      | 30.21           | 2             |
| 1649 | ORD101990 |            | CUST1173    | Andi Nugroho      | Midwest | Sports & Outdoors | Resistance Bands | 2        | 32.18      | Completed    | Debit Card     | 32.18           | 2             |
| 1670 | ORD101307 |            | CUST1248    | Andi Pratama      | East    | Toys              | Board Game       | 1        | 32.42      | Completed    | Credit Card    | 32.42           | 1             |
| 1833 | ORD100317 |            | CUST1674    | Dewi Lee          | East    | Sports & Outdoors | Resistance Bands | 1        | 35.10      | Completed    | PayPal         | 35.10           | 1             |
| 1940 | ORD102270 |            | CUST1093    | Kevin Nugroho     | Midwest | Electronics       | Smartphone Case  | 1        | 37.47      | Completed    | PayPal         | 37.47           | 1             |
| 2167 | ORD101408 |            | CUST1808    | Michael Kurniawan | Midwest | Home & Kitchen    | Coffee Maker     | 1        | 41.66      | Pending      | Credit Card    | 41.66           | 1             |
| 2283 | ORD103651 |            | CUST1376    | Nina Yusuf        | Midwest | Toys              | Action Figure    | 2        | 44.14      | Completed    | Gift Card      | 44.14           | 2             |
| 2300 | ORD101765 |            | CUST1757    | Sarah Lee         | East    | Beauty            | Moisturizer      | 1        | 44.45      | Cancelled    | Debit Card     | 44.45           | 1             |
| 2677 | ORD103515 |            | CUST1178    | Kevin Wilson      | East    | Home & Kitchen    | Throw Pillow     | 2        | 54.89      | completed    | Gift Card      | 54.89           | 2             |
| 2707 | ORD100294 |            | CUST1462    | Michael Lee       | Midwest | Toys              | RC Car           | 4        | 55.58      | Completed    | Credit Card    | 55.58           | 4             |
| 2917 | ORD101628 |            | CUST1050    | Sarah Anderson    | East    | Sports & Outdoors | Jump Rope        | 2        | 61.56      | Completed    | Debit Card     | 61.56           | 2             |
| 2934 | ORD103310 |            | CUST1374    | Nina Garcia       | Midwest | Clothing          | Cotton T-Shirt   | 1        | 62.48      | Completed    | Credit Card    | 62.48           | 1             |
| 3108 | ORD101786 |            | CUST1379    | Putri Halim       | East    | Sports & Outdoors | Resistance Bands | 2        | 69.99      | Completed    | Debit Card     | 69.99           | 2             |
| 3113 | ORD101704 |            | CUST1198    | Nina Kurniawan    | East    | Electronics       | Smartphone Case  | 1        | 70.07      | Completed    | PayPal         | 70.07           | 1             |
| 3120 | ORD101325 |            | CUST1149    | Sarah Yusuf       | Midwest | Clothing          | Cotton T-Shirt   | 1        | 70.18      | Completed    | Gift Card      | 70.18           | 1             |
| 3167 | ORD102122 |            | CUST1178    | David Salim       | East    | Sports & Outdoors | Camping Tent     | 2        | 72.10      | Completed    | PayPal         | 72.10           | 2             |
| 3260 | ORD101426 |            | CUST1278    | David Salim       | East    | Electronics       | USB-C Cable      | 1        | 76.10      | Pending      | PayPal         | 76.10           | 1             |
| 3301 | ORD101644 |            | CUST1174    | Maria Wijaya      | East    | Home & Kitchen    | Non-Stick Pan    | 1        | 77.96      | completed    | Credit Card    | 77.96           | 1             |
| 3377 | ORD100056 |            | CUST1415    | Sarah Salim       | East    | Sports & Outdoors | Hiking Backpack  | 1        | 82.78      | Completed    | Credit Card    | 82.78           | 1             |
| 3516 | ORD101650 |            | CUST1618    | Putri Wijaya      | Midwest | Electronics       | Power Bank       | 1        | 91.79      | Completed    | PayPal         | 91.79           | 1             |
| 3551 | ORD100014 |            | CUST1590    | Putri Tan         | East    | Home & Kitchen    | LED Desk Lamp    | 3        | 93.60      | Completed    | Gift Card      | 93.60           | 3             |
| 3573 | ORD100037 |            | CUST1806    | Ahmad Hidayat     | East    | Electronics       | Wireless Mouse   | 1        | 94.76      | Completed    | PayPal         | 94.76           | 1             |
| 3618 | ORD101425 |            | CUST1238    | John Pratama      | East    | Electronics       | Wireless Earbuds | 2        | 97.63      | Completed    | Debit Card     | 97.63           | 2             |
| 3706 | ORD102525 |            | CUST1233    | Michael Lee       | East    | Electronics       | Smartphone Case  | 2        | 104.81     | Returned     | Debit Card     | 104.81          | 2             |
| 3734 | ORD102969 |            | CUST1816    | John Kurniawan    | East    | Home & Kitchen    | Coffee Maker     | 1        | 107.95     | Completed    | Gift Card      | 107.95          | 1             |
| 3738 | ORD103137 |            | CUST1086    | Rina Anderson     | West    | Home & Kitchen    | Non-Stick Pan    | 2        | 108.13     | Pending      | PayPal         | 108.13          | 2             |
| 3778 | ORD102975 |            | CUST1246    | Ayu Brown         | East    | Electronics       | Laptop Stand     | 1        | 112.19     | Completed    | Debit Card     | 112.19          | 1             |
| 3868 | ORD102487 |            | CUST1143    | Linda Wijaya      | East    | Electronics       | Laptop Stand     | 1        | 129.86     | Completed    | PayPal         | 129.86          | 1             |
| 3916 |           |            |             |                   |         |                   |                  |          |            |              |                |                 |               |

Gambar 31. Kolom *region\_recovered* diurutkan A-Z untuk mempermudah penyalinan hasil per grup ke kolom asli

- *order\_date*: 40 baris kosong dihapus karena tanggal transaksi krusial untuk analisis tren dan tidak bisa direkonstruksi secara valid.

### 3.6 Pembuatan Calculated Column (Revenue)

Kolom revenue ditambahkan sebagai perkalian sederhana antara *quantity* dan *unit\_price*, sehingga nilainya otomatis konsisten dengan seluruh proses cleaning sebelumnya:

```
=[@quantity]*[@unit_price]
```

| H        | I          | J            | K              | L             | M       |
|----------|------------|--------------|----------------|---------------|---------|
| quantity | unit_price | order_status | payment_method | quantity_fill | revenue |
| 1        | 6.00       | Completed    | Gift Card      | 1             | 6.00    |
| 1        | 6.22       | Completed    | Gift Card      | 1             | 6.22    |
| 2        | 6.22       | Completed    | Credit Card    | 2             | 12.44   |
| 2        | 6.27       | Completed    | Debit Card     | 2             | 12.54   |
| 1        | 6.38       | Completed    | Gift Card      | 1             | 6.38    |
| 4        | 6.40       | Completed    | Debit Card     | 4             | 25.60   |
| 2        | 6.72       | Completed    | Gift Card      | 2             | 13.44   |
| 1        | 6.80       | Returned     | PayPal         | 1             | 6.80    |
| 2        | 6.84       | Returned     | PayPal         | 2             | 13.68   |
| 2        | 6.86       | Completed    | Credit Card    | 2             | 13.72   |

Gambar 32. Kolom *revenue* pada baris-baris awal setelah formula diterapkan

| H        | I          | J            | K              | L             | M       | N | O |
|----------|------------|--------------|----------------|---------------|---------|---|---|
| quantity | unit_price | order_status | payment_method | quantity_fill | revenue |   |   |
| 1        | 6.00       | Completed    | Gift Card      | 1             | 6.00    |   |   |
| 1        | 6.22       | Completed    | Gift Card      | 1             | 6.22    |   |   |
| 2        | 6.22       | Completed    | Credit Card    | 2             | 12.44   |   |   |
| 2        | 6.27       | Completed    | Debit Card     | 2             | 12.54   |   |   |
| 1        | 6.38       | Completed    | Gift Card      | 1             | 6.38    |   |   |
| 4        | 6.40       | Completed    | Debit Card     | 4             | 25.60   |   |   |
| 2        | 6.72       | Completed    | Gift Card      | 2             | 13.44   |   |   |
| 1        | 6.80       | Returned     | PayPal         | 1             | 6.80    |   |   |
| 2        | 6.84       | Returned     | PayPal         | 2             | 13.68   |   |   |
| 2        | 6.86       | Completed    | Credit Card    | 2             | 13.72   |   |   |
| 1        | 6.92       | Completed    | PayPal         | 1             | 6.92    |   |   |
| 2        | 6.95       | Completed    | Debit Card     | 2             | 13.90   |   |   |
| 1        | 6.96       | Completed    | Debit Card     | 1             | 6.96    |   |   |
| 1        | 6.98       | Completed    | Gift Card      | 1             | 6.98    |   |   |
| 1        | 6.98       | Completed    | Gift Card      | 1             | 6.98    |   |   |
| 1        | 7.06       | Pending      | Credit Card    | 1             | 7.06    |   |   |
| 2        | 7.18       | Completed    | PayPal         | 2             | 14.36   |   |   |
| 4        | 7.19       | Completed    | Gift Card      | 4             | 28.76   |   |   |
| 1        | 7.25       | Completed    | Gift Card      | 1             | 7.25    |   |   |
| 1        | 7.26       | Completed    | Gift Card      | 1             | 7.26    |   |   |
| 4        | 7.44       | Completed    | PayPal         | 4             | 29.76   |   |   |
| 2        | 7.83       | Completed    | PayPal         | 2             | 15.66   |   |   |
| 1        | 7.84       | Completed    | Gift Card      | 1             | 7.84    |   |   |
| 1        | 7.88       | Completed    | Debit Card     | 1             | 7.88    |   |   |
| 5        | 7.95       | Completed    | Credit Card    | 5             | 39.75   |   |   |
| 1        | 8.13       | Pending      | Debit Card     | 1             | 8.13    |   |   |
| 2        | 8.17       | Completed    | Credit Card    | 2             | 16.34   |   |   |
| 3        | 8.20       | Completed    | Debit Card     | 3             | 24.60   |   |   |
| 2        | 8.22       | Completed    | Debit Card     | 2             | 16.44   |   |   |
| 3        | 8.23       | Cancelled    | Gift Card      | 3             | 24.69   |   |   |
| 1        | 8.29       | Completed    | Gift Card      | 1             | 8.29    |   |   |
| 1        | 8.29       | Completed    | Debit Card     | 1             | 8.29    |   |   |
| 1        | 8.33       | Completed    | Gift Card      | 1             | 8.33    |   |   |
| 1        | 8.39       | Completed    | Credit Card    | 1             | 8.39    |   |   |
| 1        | 8.41       | Completed    | Gift Card      | 1             | 8.41    |   |   |
| 1        | 8.41       | Completed    | Credit Card    | 1             | 8.41    |   |   |
| 1        | 8.41       | Pending      | Credit Card    | 1             | 8.41    |   |   |
| 1        | 8.43       | Completed    | Debit Card     | 1             | 8.43    |   |   |
| 3        | 8.47       | Returned     | Credit Card    | 3             | 25.41   |   |   |
| 1        | 8.49       | Completed    | Gift Card      | 1             | 8.49    |   |   |
| 1        | 8.56       | Pending      | Gift Card      | 1             | 8.56    |   |   |
| 1        | 8.59       | Completed    | PayPal         | 1             | 8.59    |   |   |
| 3        | 8.64       | Completed    | PayPal         | 3             | 25.92   |   |   |
| 1        | 8.75       | Completed    | Debit Card     | 1             | 8.75    |   |   |
| 1        | 8.75       | Completed    | Gift Card      | 1             | 8.75    |   |   |
| 1        | 8.87       | Completed    | Credit Card    | 1             | 8.87    |   |   |
| 1        | 8.88       | Completed    | Gift Card      | 1             | 8.88    |   |   |
| 1        | 8.90       | Completed    | PayPal         | 1             | 8.90    |   |   |
| 1        | 8.91       | Pending      | Credit Card    | 1             | 8.91    |   |   |
| 2        | 8.97       | Completed    | Gift Card      | 2             | 17.94   |   |   |
| 2        | 8.98       | Completed    | Gift Card      | 2             | 17.96   |   |   |
| 1        | 9.04       | Completed    | Gift Card      | 1             | 9.04    |   |   |
| 1        | 9.04       | Completed    | Credit Card    | 1             | 9.04    |   |   |
| 2        | 9.04       | Completed    | Debit Card     | 2             | 18.08   |   |   |
| 1        | 9.05       | Completed    | Credit Card    | 1             | 9.05    |   |   |
| 4        | 9.05       | Completed    | PayPal         | 4             | 36.20   |   |   |
| 1        | 9.05       | Completed    | Unknown        | 1             | 9.05    |   |   |
| 1        | 9.06       | Completed    | Debit Card     | 1             | 9.06    |   |   |
| 1        | 9.06       | Completed    | PayPal         | 1             | 9.06    |   |   |
| 3        | 9.07       | Completed    | Debit Card     | 3             | 27.21   |   |   |
| 1        | 9.08       | Completed    | PayPal         | 1             | 9.08    |   |   |

Gambar 33. Kolom *revenue* pada dataset — total revenue \$306.888,66 dari 3.934 baris (sebelum 1 duplikat tambahan ditemukan pada verifikasi ulang di bagian 3.1)

| Hari  | Fase                | Task                                            | Tools | Selesai? |
|-------|---------------------|-------------------------------------------------|-------|----------|
| Day 1 | Excel Data Cleaning | Ambil dataset e-commerce sales                  | Excel | SELESAI  |
| Day 1 | Excel Data Cleaning | Cek & tangani missing values                    | Excel | SELESAI  |
| Day 1 | Excel Data Cleaning | Cek & hapus duplicate records                   | Excel | SELESAI  |
| Day 1 | Excel Data Cleaning | Perbaiki data types yang salah                  | Excel | SELESAI  |
| Day 1 | Excel Data Cleaning | Standarkan kategori yang tidak konsisten        | Excel | SELESAI  |
| Day 1 | Excel Data Cleaning | Format ulang kolom tanggal                      | Excel | SELESAI  |
| Day 1 | Excel Data Cleaning | Cek outlier                                     | Excel | SELESAI  |
| Day 1 | Excel Data Cleaning | Buat calculated columns                         | Excel | SELESAI  |
| Day 1 | Excel Data Cleaning | Dokumentasikan raw -> cleaning -> clean dataset | Excel |          |
| Day 1 | Excel Data Cleaning | Simpan sebagai cleaned_sales_data.xlsx          | Excel |          |

Gambar 34. Checklist Day 1 — seluruh tahap cleaning selesai, tinggal dokumentasi dan penyimpanan file

## 4. Ringkasan Hasil (Cleaning Log)

Tabel berikut merangkum seluruh langkah cleaning secara berurutan, dan disertakan sebagai sheet terpisah ("Cleaning Log") dalam file cleaned\_sales\_data.xlsx sebagai jejak audit proses.

| Langkah                            | Baris Terdampak | Keterangan                                                                                                                                                                             |
|------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Hapus duplicate records         | 99 baris        | Exact duplicate (seluruh kolom identik), diduga input ganda — 98 baris pada pembersihan awal + 1 baris ditemukan saat verifikasi ulang di Day 2 (teknik row signature + filter warna). |
| 2. Standardisasi category & region | —               | Diseragamkan ke Proper Case; typo diperbaiki (Cloths→Clothing, Electronic→Electronics, Mid-West→Midwest).                                                                              |
| 3. Standardisasi order_date        | —               | 5 format tanggal disatukan menjadi YYYY-MM-DD.                                                                                                                                         |
| 4. Hapus outlier quantity          | 15 baris        | Nilai ekstrem (250/500/999) tidak masuk akal untuk transaksi retail individual, tersebar random tanpa pola wholesale.                                                                  |
| 5. Hapus outlier unit_price        | 12 baris        | Nilai ekstrem (8888/4999,99/1234,56) tidak sesuai kategori produk; nilai identik berulang mengindikasikan error sistem.                                                                |
| 6. Tangani missing value           | 533 sel         | Lihat Tabel 2 — strategi berbeda per kolom sesuai tingkat kepentingan & kemungkinan recovery.                                                                                          |
| 7. Buat calculated column revenue  | 3.933 baris     | revenue = quantity × unit_price. Total revenue dataset: \$306.888,66.                                                                                                                  |

Tabel 3. Cleaning Log — ringkasan seluruh proses raw dataset → clean dataset

## 5. Kesimpulan

Dataset akhir berisi 3.933 baris bersih, tanpa missing value, dengan tipe data dan format yang konsisten pada seluruh kolom. Total 167 baris (4,1% dari data awal) dihapus sepanjang proses, seluruhnya melalui keputusan yang terdokumentasi dan dapat dipertanggungjawabkan, bukan penghapusan sepihak tanpa alasan. Satu duplikat tambahan yang lolos dari pembersihan awal berhasil ditemukan lewat verifikasi ulang sebelum data dipakai di tahap analisis — praktik quality assurance yang penting dilakukan sebelum data dianggap final.

Output tahap ini adalah file cleaned\_sales\_data.xlsx yang berisi dua sheet: data transaksi yang sudah bersih, dan Cleaning Log sebagai dokumentasi proses. Dataset ini menjadi sumber tunggal yang dipakai secara konsisten pada tahap-tahap berikutnya: analisis Excel (Day 2), query SQL (Day 3), exploratory data analysis dengan Python (Day 4), dan dashboard Tableau (Day 5).